

Technical Project:

VMware Virtualization Home Lab

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Summary

This project aims to build a **local virtualization lab** to practice real-world **system administration tasks** using **VMware vSphere**.

The goal is to simulate a **professional IT infrastructure environment** from scratch, in order to understand how resources, services, and dependencies are managed in a production setting.

The lab is based on **VMware ESXi 8.0** and **vCenter Server Appliance (VCSA)**, both deployed in a **nested environment** within **VMware Workstation Pro 17.5**.

On top of this infrastructure, several test virtual machines are created and managed, performing common operations such as **creation, snapshots, cloning, resizing, and tagging**, replicating the typical daily work of a **platform or systems team**.

Beyond the technical component, the purpose of this project is to produce comprehensive, structured, and verifiable documentation, serving both as a **personal learning resource** and as **evidence of technical capability** and readiness to progress toward more specialized roles in systems administration.

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1. Objectives and Technical Scope

1.1 General Objective

The purpose of this project is to build a complete VMware virtualization lab that replicates, in a controlled local environment, the standard operations of an enterprise infrastructure.

The main goal is to develop real-world skills in virtualized systems administration — understanding vSphere architecture, deploying the ESXi hypervisor, configuring networking, DNS and NTP services, and installing the vCenter Server Appliance (VCSA).

This lab is designed as a practical and educational foundation for transitioning toward a more advanced technical role oriented to **platform, systems, or cloud administration**.

1.2 Specific Objectives

- Deploy a **VMware ESXi 8.0** host on **VMware Workstation Pro 17** in a nested environment.
- Configure the environment's core services: **NAT networking**, **NTP**, and **internal DNS** (Windows Server 2022).
- Deploy and configure **vCenter Server Appliance (VCSA 8.0)**.
- Validate **connectivity** and **FQDN resolution** via **nslookup**, **ping**, and vSphere Client access.
- Create two test virtual machines — Windows Server and Ubuntu Server — and manage their lifecycle.
- Perform realistic administrative operations:
 - Creation, cloning, and snapshotting of VMs.
 - Resource resizing.
 - Configuration of alarms, tags, and metrics in vSphere.
- Document every process with screenshots, logs, and verifiable results.
- Establish the foundation for the following **Veeam Backup & Replication** project, ensuring technical compatibility and continuity.

1.3 Project Scope

The scope includes the complete construction of the lab, its network and service configuration, and comprehensive technical documentation of each phase.

Included:

- Implementation of a functional environment (ESXi, DNS, NTP).
- Deployment attempts and troubleshooting of VCSA.
- Validation of connectivity, DNS, and time synchronization.

Not included:

- Public-cloud integrations.

The focus remains on reproducing a **fully functional and realistic environment** within **limited hardware resources**.

1.4 Professional Context

This project is part of a technical and professional development process aimed at transitioning into a more specialized IT role.

All work was performed outside regular working hours, using personal resources, and following official VMware documentation and community guides.

The main motivation was to understand the *technical reasoning* behind each step — not merely to replicate procedures — and to produce documentation that demonstrates **technical judgement and real troubleshooting ability**.

2. Environment Design and Architecture

2.1 General Structure

The lab was built entirely on a single physical machine acting as the main host through **VMware Workstation Pro 17**.

Within it, a **VMware ESXi 8.0** virtual host was deployed, hosting the remaining components:

- **vCenter Server Appliance (VCSA)** — centralized management.
- **Windows Server 2022** — internal DNS role.
- **Ubuntu Server 22.04** — Linux test machine.

The environment is isolated within a **local NAT network (VMnet8)**, simulating a compact, self-contained corporate infrastructure.

2.2 Lab Topology

Element	Role	Operating System / Version	Assigned Resources	IP Address	Notes
Physical Host	Lab base	Windows 11 Pro + VMware Workstation 17	32 GB RAM / i7-12700KF / 1 TB SSD	DHCP	Nested support
VMware ESXi	Primary hypervisor	VMware ESXi 8.0	4 vCPU / 8 GB RAM / 40 GB disk	192.168.119.150	Host Client
SRV-DN S01	Internal DNS	Windows Server 2022 Eval.	2 vCPU / 4 GB RAM / 40 GB disk	192.168.119.160	DNS Server role
SRV-VC SA	Centralized management	vCenter Server Appliance 8.0	4 vCPU / 14 GB RAM / 530 GB disk	192.168.119.170	Partial deployment

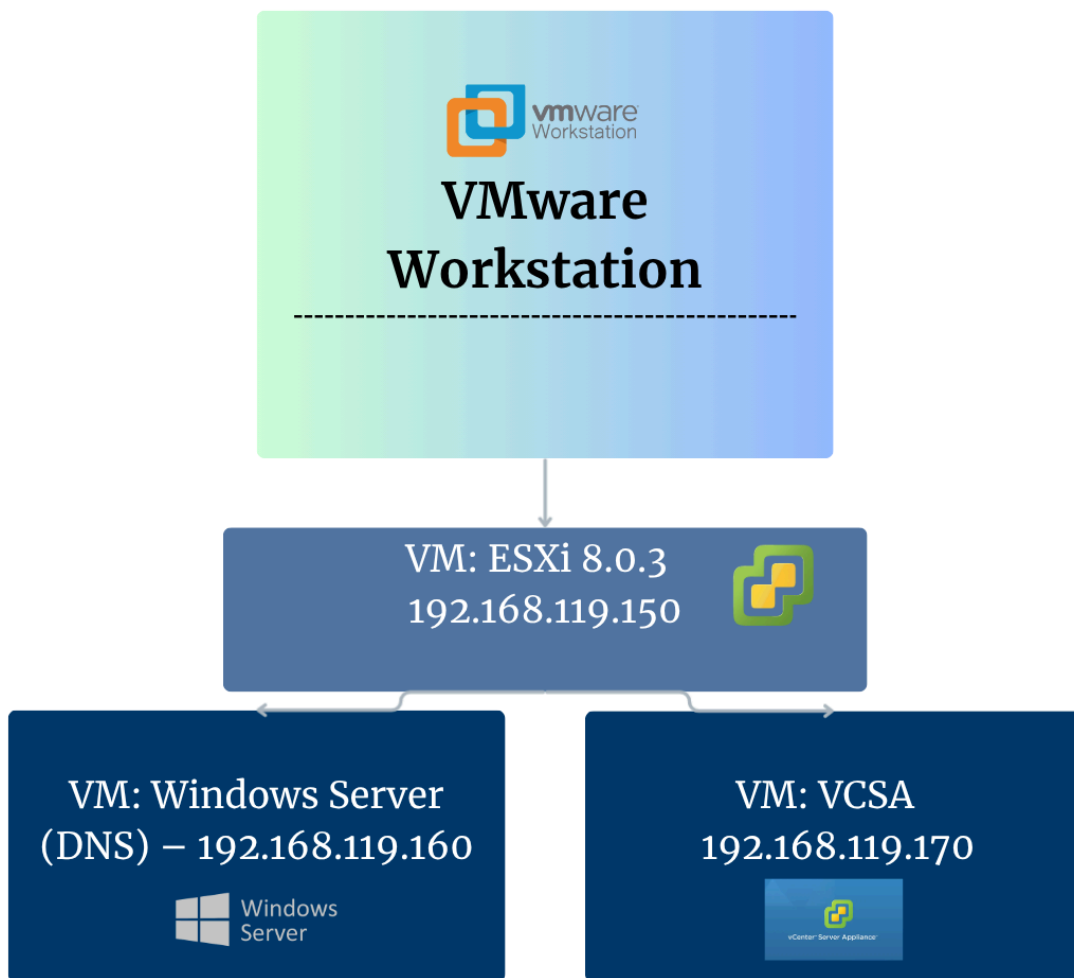
2.3 Networking and Connectivity

- **Network type:** VMnet8 (NAT) — provides Internet access and lab isolation.
- **NAT gateway:** 192.168.119.2
- **Network range:** 192.168.119.0/24
- **Internal DNS:** 192.168.119.160 (SRV-DNS01)
- All systems (ESXi, DNS, VCSA and clients) communicate within the same subnet.
- FQDN and PTR resolution validated: [vcenter.lab.local](#) ↔ [192.168.119.170](#).
- Time synchronization established with public NTP servers ([time.google.com](#)), verified via `ntpq -p`.

2.4 Environment Resources

Resource	Value / Configuration
Physical CPU	Intel Core i7-12700KF (12 C / 20 T)
Memory	32 GB DDR4
Total storage	1 TB NVMe SSD
Allocated to lab	~550 GB (thin-provisioned)
Base hypervisor	VMware Workstation 17
Nested hypervisor	VMware ESXi 8.0
Virtual network	VMnet8 (NAT)
Minimum per VM	2–4 vCPU / 14 GB RAM / 530 GB disk (VCSA Tiny)
Admin client OS	Windows 11

2.5 Environment Diagram



3. Requirements and Resources

3.1 Hardware Requirements

Component	Minimum Requirement	Actual Configuration
CPU	Processor with VT-x / AMD-V enabled	Intel Core i7-12700KF
RAM	24 GB (minimum) / 32 GB recommended	32 GB DDR4
Storage	500 GB free / SSD recommended	1 TB NVMe SSD
Network	Adapter compatible with NAT/Bridge	VMnet8 in Workstation
GPU	Optional, non-critical	Integrated

3.2 Software Requirements

Software	Version	Primary Use
VMware Workstation Pro	17.5+	Base virtualization (physical host)
VMware ESXi	8.0	Nested hypervisor
vCenter Server Appliance	8.0	Centralized management
Windows Server	2022 Eval.	Internal DNS server
OVF Tool	5.0.0	OVA → OVF conversion
7-Zip, Notepad++	—	Extraction / OVF editing
Public NTP	pool.ntp.org	Time synchronization

3.3 Dependencies and Considerations

- **DNS is a critical prerequisite:** the VCSA installer requires both forward and reverse resolution of the FQDN ([vcenter.lab.local](#)).
- **Clock synchronization** between ESXi, DNS, and the client must be ensured via NTP.

- The **VMnet8 (NAT)** network provides Internet access while keeping the lab isolated.
- The **VCSA deployment** requires at least 14 GB of RAM, limiting available host resources.
- All services are **fully local**, with no dependency on public cloud infrastructure.
- All steps followed **VMware 8.0 official best practices** (Installation & Setup Guide).

4. Environment Setup — Step by Step

4.1 ESXi Deployment and Installation

4.1.1 Objective

Install and bring online the **VMware ESXi 8.0** hypervisor for the lab — network-accessible with a static IP, web and SSH access enabled, and a datastore prepared to host the virtual machines (including the upcoming VCSA deployment).

4.1.2 Actions Performed

Creation of the ESXi VM (nested in VMware Workstation):

- CPU: 4 vCPU (Intel/AMD compatible depending on host).
- RAM: 16 GB (8 GB minimum functional).
- System disk: 8–16 GB (thin-provisioned).
- Datastore disk: ≥ 600 GB (thin-provisioned).
- NIC: VMnet8 (NAT) → lab network [192.168.119.0/24](#).
- Mounted ESXi 8.x ISO and booted from CD.

Base installation of the hypervisor:

- Standard installation accepting the EULA and selecting the system disk.
- Created **root** user with a secure password.
- Verified **DCUI console** on first boot.

Management Network configuration (via DCUI):

From **F2** → **Configure Management Network**:

- IPv4 static: [192.168.119.150](#)
- Netmask: [255.255.255.0](#)
- Gateway: [192.168.119.2](#) (Workstation NAT gateway)

- Temporary DNS: [8.8.8.8](#) (later replaced by local DNS)
- Hostname: [esxi.lab.local](#)
- Visual confirmation in DCUI of IP and connectivity.

Basic setup and initial validations:

- Browser access: <https://192.168.119.150/> → **ESXi Host Client**.
- Temporary SSH enabled under *Manage* → *Services*.
- License verified (trial version).
- Confirmed datastore ≥ 600 GB and **vSwitch0** bound to VMnet8 NIC.

Validations from the host PC:

- Successful ping to ESXi IP.

SSH connection to host:

```
$ ssh root@192.168.119.150
```

4.1.3 Issues Found

Network not reachable after setting static IP

Cause: VM was not attached to VMnet8 or gateway misconfigured.

Solution: Force VMnet8 and gateway [192.168.119.2](#).

Subnet confusion (192.168.1.150 vs 192.168.119.150)

Cause: Initial config outside NAT range.

Solution: Standardized all devices to [192.168.119.0/24](#).

Insufficient datastore space

Cause: VCSA Tiny requires > 579 GB provisioned storage.

Solution: Added a thin disk ≥ 600 GB for VMs.

Limited Internet access (NTP/DNS)

Cause: NAT or DNS not resolving properly.

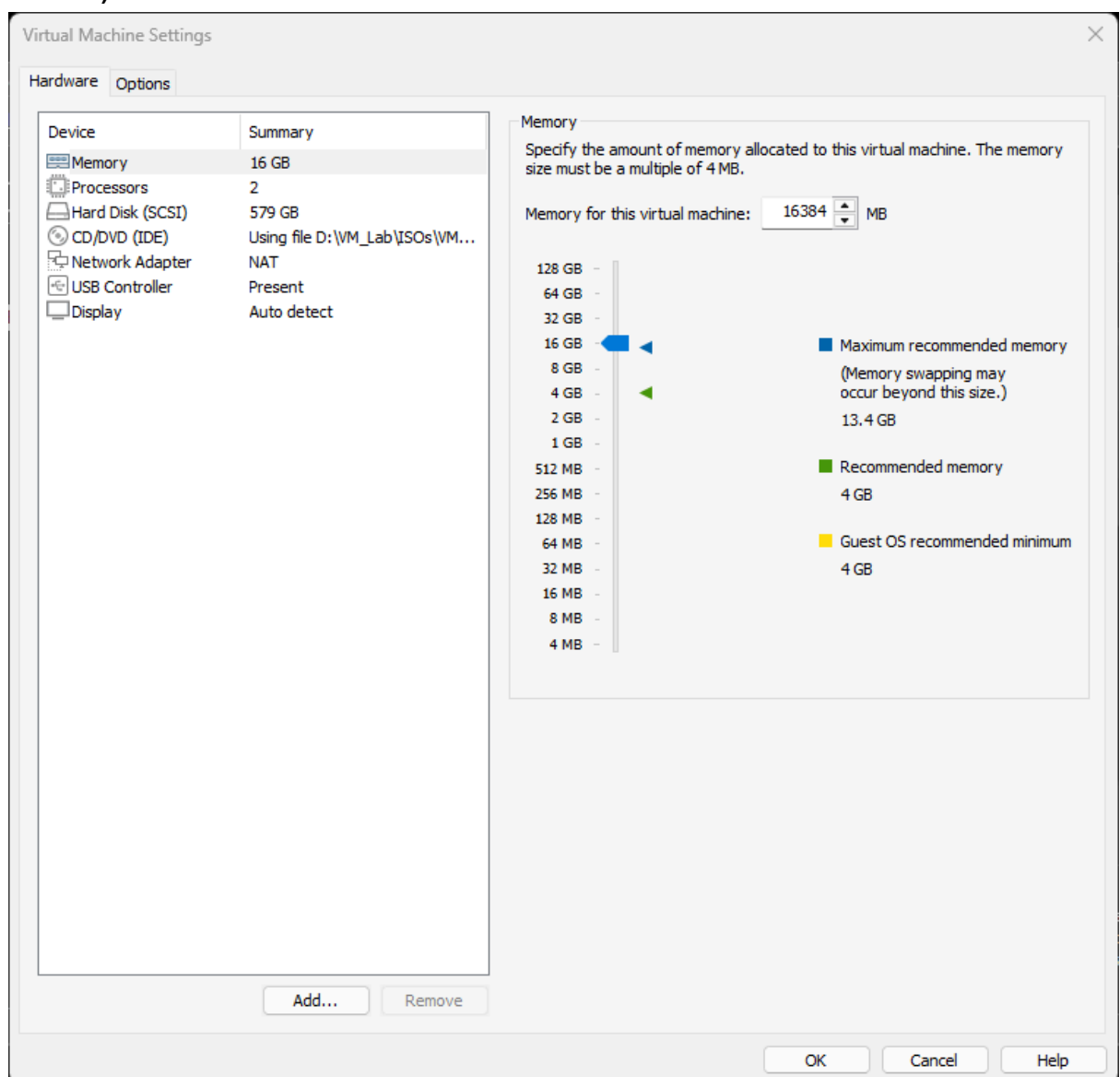
Solution: Temporarily used [8.8.8.8](#) until local DNS was set up.

4.1.4 Final Results

- ESXi operational and reachable via **HTTPS** and **SSH**.
- Static IP **192.168.119.150** correctly configured.
- Active datastore with sufficient capacity.
- Environment ready for configuration of **NTP (4.2)**, **DNS (4.3)**, and **VCSA (4.4)**.

4.1.5 Evidence

- **Figure 4.1.1 – ESXi VM configuration in Workstation (CPU/RAM/Disks/NIC → VMnet8).**



- Figure 4.1.2 – DCUI: *Configure Management Network* with static IP and gateway 192.168.119.2.

IPv4 Configuration

This host can obtain network settings automatically if your network includes a DHCP server. If it does not, the following settings must be specified:

() Disable IPv4 configuration for management network

() Use dynamic IPv4 address and network configuration

(o) Set static IPv4 address and network configuration:

IPv4 Address

Subnet Mask

Default Gateway

[192.168.119.150]

[255.255.255.0]

[192.168.119.2]

<Up/Down> Select

<Space> Mark Selected

<Enter> OK

<Esc> Cancel

- **Figure 4.1.3 – ESXi Host Client accessible at <https://192.168.119.150>.**

The screenshot displays the ESXi Host Client web interface in a browser. The address bar shows the URL <https://192.168.119.150/ui/#/host>. The interface is titled "ESXi Host Client" and shows the host "localhost.localdomain".

Actions: Obtener vCenter Server, Crear/Registrar máquina virtual, Apagar, Reiniciar, Actualizar.

Host Details:

- localhost.localdomain** (Status: Online)
- Versión: 8.0 Update 3
- Condición: Normal (no conectado a ningún vCenter)
- Tiempo en línea: 0 días

Resource Usage:

Resource	Used	Capacity	Free
CPU	154 MHz	7.2 GHz	7.1 GHz (2%)
MEMORIA	1.75 GB	16 GB	14.25 GB (11%)
ALMACENAMIENTO	91.99 GB	450.75 GB	358.76 GB (20%)

Notifications:

- Se habilitó SSH en el host localhost.localdomain
- SSH está habilitado en este host. Debería deshabilitar SSH a menos que sea necesario por razones administrativas.

Hardware Details:

Component	Details
Fabricante	VMware, Inc.
Modelo	VMware20,1
CPU	2 CPUs x 12th Gen Intel(R) Core(TM) i7-12700KF
Memoria	16 GB
Flash virtual	0 B utilizado, 0 B capacidad
Redes	- Nombre de host: localhost.localdomain - Direcciones IP: 1. vmk0: 192.168.119.150, 2. vmk0: fe80::20c:29ff:fe34:f80 - Servidores DNS: 1. 192.168.119.160 - Puerta de enlace predeterminada: 192.168.119.2 - IPv6 habilitado: Sí

Footer: Tareas recientes

- **Figure 4.1.4 – Storage → Datastores showing ≥ 600 GB free space.**

The screenshot shows the ESXi Host Client interface for a local host. The 'Almacenes de datos' (Datastores) tab is selected. The interface displays a table with one datastore, 'datastore1', which has a total capacity of 450.75 GB and 358.76 GB of free space. The interface also shows a detailed view of 'datastore1' with its type (VMFS6), location, UUID, and the number of hosts and virtual machines connected to it.

Nombre	Tipo de unidad	Capacidad	Aprovisionado	Libre	Tipo	Aprovisionar	Acceso
datastore1	No SSD	450,75 GB	91,99 GB	358,76 GB	VMFS6	Compatible	Individual

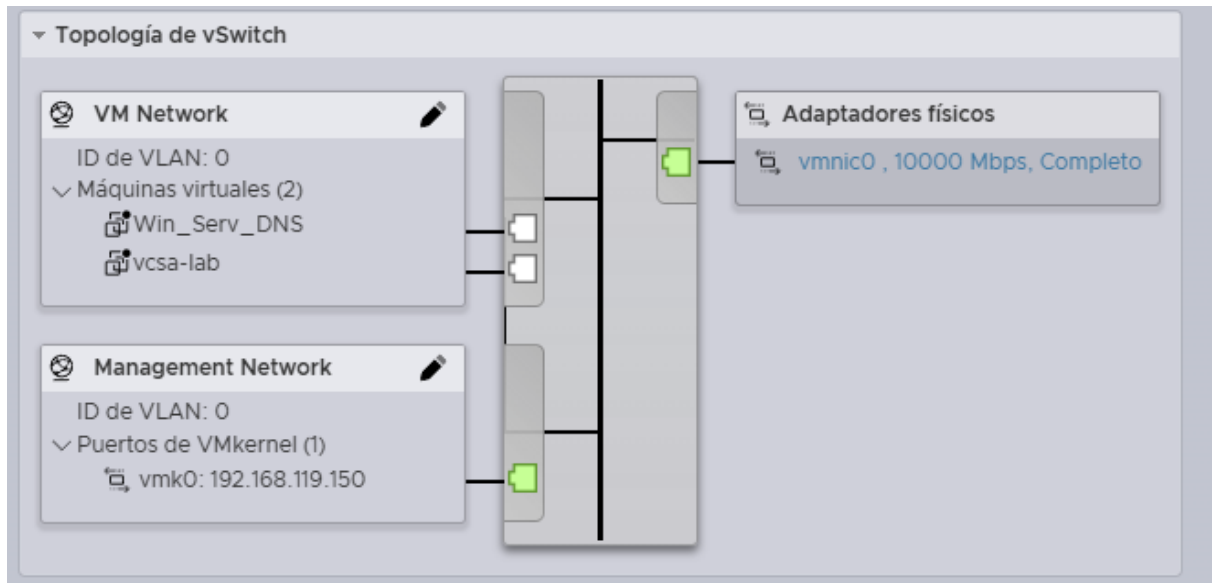
1 elementos

datastore1

Tipo: VMFS6
Ubicación: /vmfs/volumes/68b85fad-890e2d78-ef7c-000c29340f80
UUID: 68b85fad-890e2d78-ef7c-000c29340f80
Hosts: 1
Máquinas virtuales: 2

ALMACENAMIENTO LIBRE: 358,76 GB
USADO: 91,99 GB
CAPACIDAD: 450,75 GB
20%

- **Figure 4.1.5 – Networking: vSwitch0 Topology**



- **Figure 4.1.6 – Connectivity test: `ping 192.168.119.150`.**

```
PS C:\Users> ping 192.168.119.150

Haciendo ping a 192.168.119.150 con 32 bytes de datos:
Respuesta desde 192.168.119.150: bytes=32 tiempo<1m TTL=64
Respuesta desde 192.168.119.150: bytes=32 tiempo=1ms TTL=64
Respuesta desde 192.168.119.150: bytes=32 tiempo=1ms TTL=64
Respuesta desde 192.168.119.150: bytes=32 tiempo=1ms TTL=64

Estadísticas de ping para 192.168.119.150:
    Paquetes: enviados = 4, recibidos = 4, perdidos = 0
        (0% perdidos),
    Tiempos aproximados de ida y vuelta en milisegundos:
        Mínimo = 0ms, Máximo = 1ms, Media = 0ms
PS C:\Users> |
```

4.2 Network and NTP Configuration

4.2.1 Objective

Configure the ESXi host's network parameters and enable NTP time synchronization to ensure time consistency between ESXi, the DNS Server, and the VCSA.

Time synchronization is critical to prevent authentication errors and service failures within vCenter.

4.2.2 Actions performed

Management network review

- In *Networking* → *VMkernel adapters*, it was confirmed that the **Management Network** was associated with **vSwitch0** and linked to the **NIC** connected to **VMnet8 (NAT)**.
 - **Configured parameters:**
 - **IP:** 192.168.119.150
 - **Netmask:** 255.255.255.0
 - **Gateway:** 192.168.119.2
 - **Temporary DNS:** 8.8.8.8 (later replaced by the internal DNS of the lab)
- **SSH** was enabled for administration through terminal.

SSH access to the host

- Connection established from the host machine (Windows):
ssh root@192.168.119.150

NTP service configuration

- Initial NTP service status verified (stopped):
/etc/init.d/ntpd status
- NTP servers defined:
esxcli system ntp set -s 0.pool.ntp.org,1.pool.ntp.org,2.pool.ntp.org

- Service started and enabled at boot:
`/etc/init.d/ntpd start`
`esxcli system ntp set --enabled true`
- Synchronization verified: `ntpq -p`
(Figure 4.2.4)

Final network and Internet validation

- External connectivity (NAT) validated:
`ping 8.8.8.8`
- Hostname verification:
`hostname → esxi.lab.local`

4.2.3 Issues found

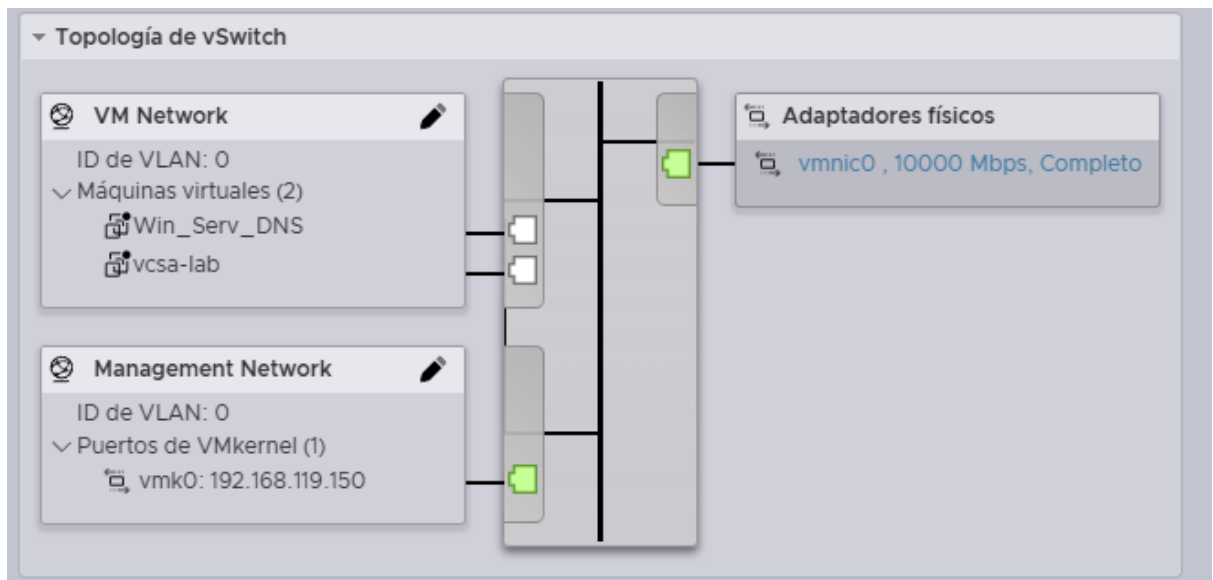
- **“Cannot resolve NTP server destination”**
 - *Cause:* The host could not resolve domains via DNS.
 - *Solution:* Manually configured DNS to `192.168.119.2` and temporarily used Google’s direct IP (`216.239.35.0`) for NTP.
- **Firewall blocking NTP**
 - *Cause:* The `ntpClient` ruleset was managed internally by the service.
 - *Solution:* Enabled NTP using `esxcli` and validated synchronization with `ntpq -p` instead of forcing rules.
- **Initial time drift**
 - *Cause:* Slight deviation between the host and lab clocks.
 - *Solution:* Synchronized and restarted the NTP service; the deviation disappeared.

4.2.4 Final results

- Management network operational with static IP [192.168.119.150](#).
- External connectivity validated through [VMnet8 \(NAT\)](#).
- NTP active and synchronized with Google's public time servers.
- Host ready for integration with internal DNS and VCSA deployment.

4.2.5 Evidence

- **Figure 4.2.1 – Networking showing Management Network on vSwitch0 → VMnet8.**



- **Figure 4.2.2 – IPv4 configuration with static IP and gateway [192.168.119.2](#).**

IPv4 Configuration

This host can obtain network settings automatically if your network includes a DHCP server. If it does not, the following settings must be specified:

☐ Disable IPv4 configuration for management network
☐ Use dynamic IPv4 address and network configuration
☒ Set static IPv4 address and network configuration:

IPv4 Address	[192.168.119.50]
Subnet Mask	[255.255.255.0]
Default Gateway	[192.168.119.2_]

<Up/Down> Select <Space> Mark Selected <Enter> OK <Esc> Cancel

- Figure 4.2.3 – SSH session to the host (`root@192.168.119.150`).

```
C:\Users>ssh root@192.168.119.150
(root@192.168.119.150) Password:
The time and date of this login have been sent to the system logs.

WARNING:
  All commands run on the ESXi shell are logged and may be included in
  support bundles. Do not provide passwords directly on the command line.
  Most tools can prompt for secrets or accept them from standard input.

VMware offers powerful and supported automation tools. Please
see https://developer.vmware.com for details.

The ESXi Shell can be disabled by an administrative user. See the
vSphere Security documentation for more information.
[root@localhost:~] |
```

- Figure 4.2.4 – `ntpq -p` output showing reachable peer `time1.google.com`.

```
[root@localhost:~] ntpq -p
      remote           refid      st t when poll reach  delay  offset  jitter
=====
*216.239.35.0    .GOOG.          1 u  33   64  377   29.411   +6.944   0.513
[root@localhost:~] |
```

- Figure 4.2.5 – Successful `ping 8.8.8.8` from ESXi console.

```
[root@localhost:~] ping 8.8.8.8
PING 8.8.8.8 (8.8.8.8): 56 data bytes
64 bytes from 8.8.8.8: icmp_seq=0 ttl=128 time=10.824 ms
64 bytes from 8.8.8.8: icmp_seq=1 ttl=128 time=10.803 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=128 time=10.256 ms

--- 8.8.8.8 ping statistics ---
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 10.256/10.628/10.824 ms

[root@localhost:~] |
```

4.3 Implementation of the DNS Server (Windows Server)

4.3.1 Objective

Deploy and configure an internal DNS server (**SRV-DNS01**) within the virtualized environment to manage name resolution under the **lab.local** domain.

The goal is to provide forward and reverse resolution for all components in the lab (ESXi, VCSA, and the DNS server itself), ensuring that FQDNs are properly recognized and verified during the **vCenter Server Appliance (VCSA)** deployment.

Note: Windows Server 2022 Standard (Evaluation) was used for its ease of graphical administration and full compatibility with VMware environments.

Perfecto. Ya tengo claro el formato que quieres (listas con viñetas, sin bloques de código, estilo igual al documento en castellano).

Aquí tienes **4.3.2 Actions performed** completamente ajustado al formato correcto:

4.3.2 Actions performed

Creation of the virtual machine SRV-DNS01

- **Name:** SRV-DNS01
- **Operating system:** Windows Server 2022 Standard (Evaluation)
- **Assigned configuration:**
 - 2 vCPU
 - 4 GB RAM
 - 80 GB HDD (thin provisioned)
 - Network adapter: VMnet8 (NAT)
- **Static IP configuration:**
 - IP: 192.168.119.160
 - Netmask: 255.255.255.0
 - Gateway: 192.168.119.2
 - Preferred DNS: 192.168.119.160 (self-reference)

- Secondary DNS: 8.8.8.8

DNS role installation

- Accessed *Server Manager* → *Add Roles and Features*.
- Selected the **DNS Server** role without any additional features.
- Completed the setup and verified the installation in *Server Manager* → *Tools* → *DNS*.
- Confirmed that the **DNS service** started automatically after installation.

DNS zone configuration

- **Forward lookup zone:**
 - Name: lab.local
 - Type: Primary zone
 - Storage: Local on the server
- **A records created:**
 - esxi.lab.local → 192.168.119.150 (Main ESXi host)
 - vcenter.lab.local → 192.168.119.170 (Future VCSA)
 - dns.lab.local → 192.168.119.160 (Internal DNS server)
- **Reverse lookup zone:**
 - Name: 119.168.192.in-addr.arpa
 - Type: Primary zone
 - Automatic creation of PTR records linked to the above A records.

DNS resolution verification

- From the server console (CMD):
 - nslookup vcenter.lab.local

- `nslookup 192.168.119.170`
- **Expected result:**
 - Name: vcenter.lab.local → Address: 192.168.119.170
 - Name: 170.119.168.192.in-addr.arpa → Address: vcenter.lab.local
- Verified that the server correctly resolved both by name and by IP.
- The DNS server successfully resolved itself (127.0.0.1 and 192.168.119.160) and used 8.8.8.8 as a secondary forwarder for external queries.

4.3.3 Issues found

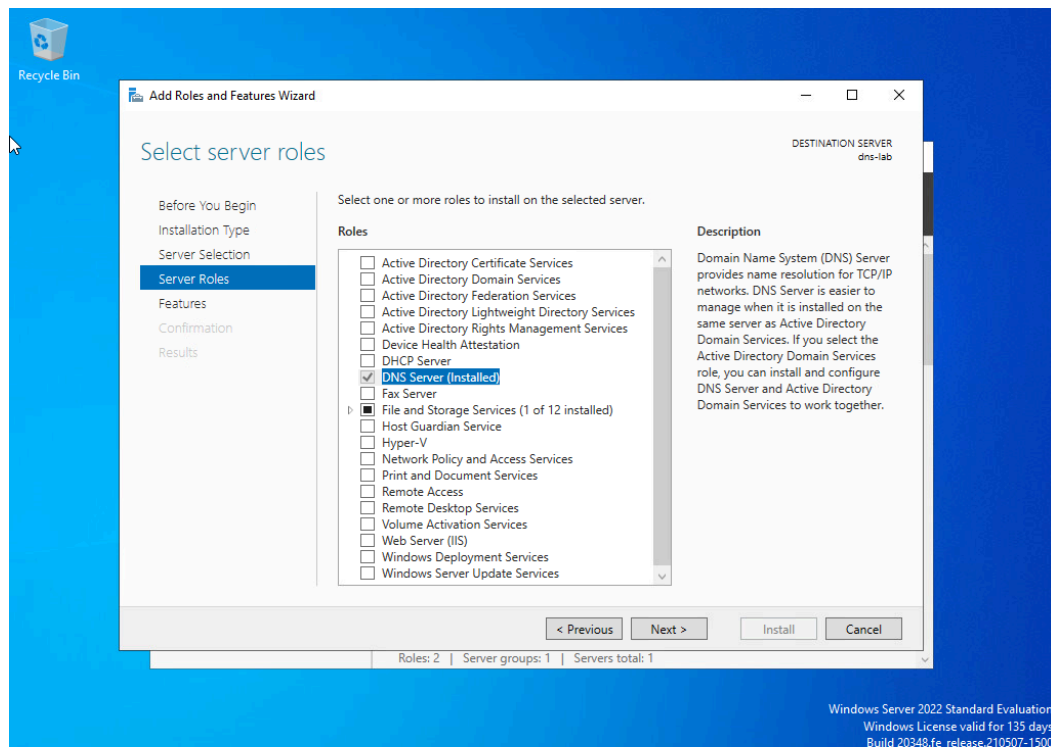
- **Initial nslookup failure**
 - *Cause:* The preferred DNS was set to 8.8.8.8 instead of its own IP.
 - *Solution:* Set 192.168.119.160 as the primary DNS in the network interface.
- **Reverse resolution failure**
 - *Cause:* The reverse lookup zone was created with an incorrect prefix.
 - *Solution:* Recreate the zone 119.168.192.in-addr.arpa to allow proper PTR resolution.
- **Nonexistent FQDN for vCenter (not yet deployed)**
 - *Cause:* The record vcenter.lab.local did not correspond to an active host.
 - *Solution:* Keep the A record ready until VCSA deployment and verify afterward.

4.3.4 Final results

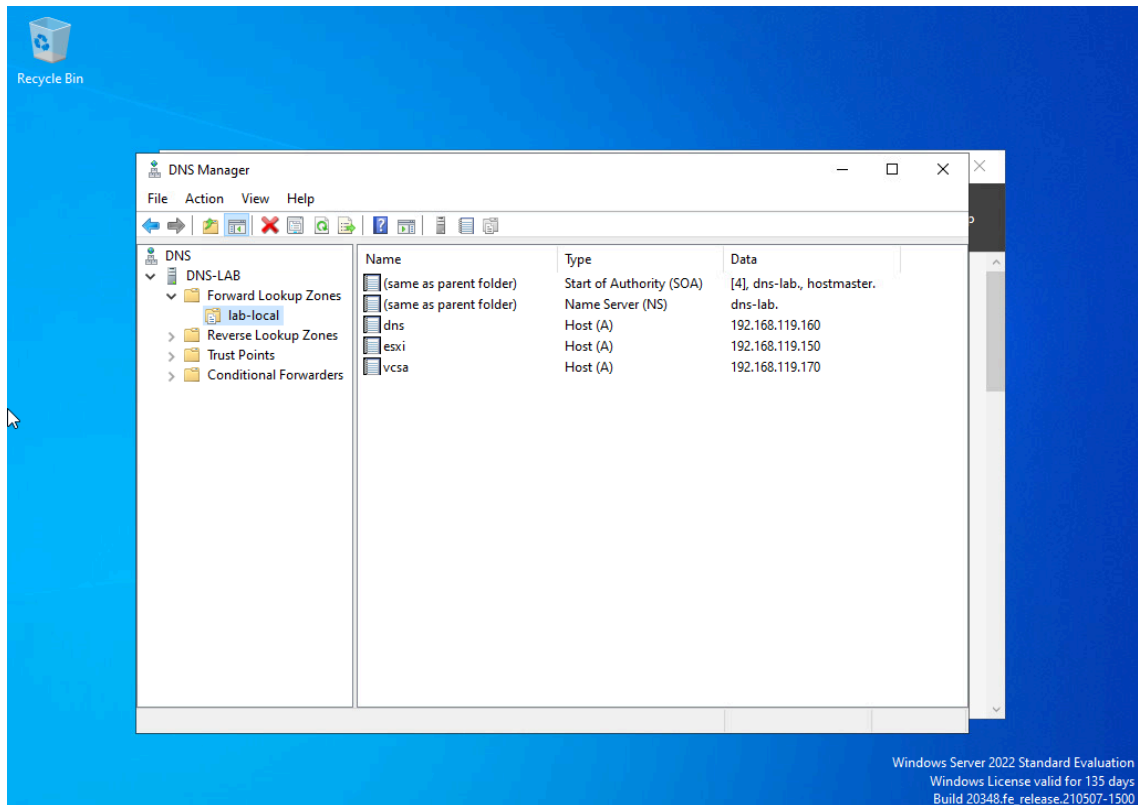
- Internal DNS server operational and correctly resolving all defined records.
- Communication established with ESXi and Internet access through NAT.
- **lab.local** domain fully functional with both forward and reverse resolution.
- Ready to serve as the central FQDN resolver during VCSA installation.

4.3.5 Evidence

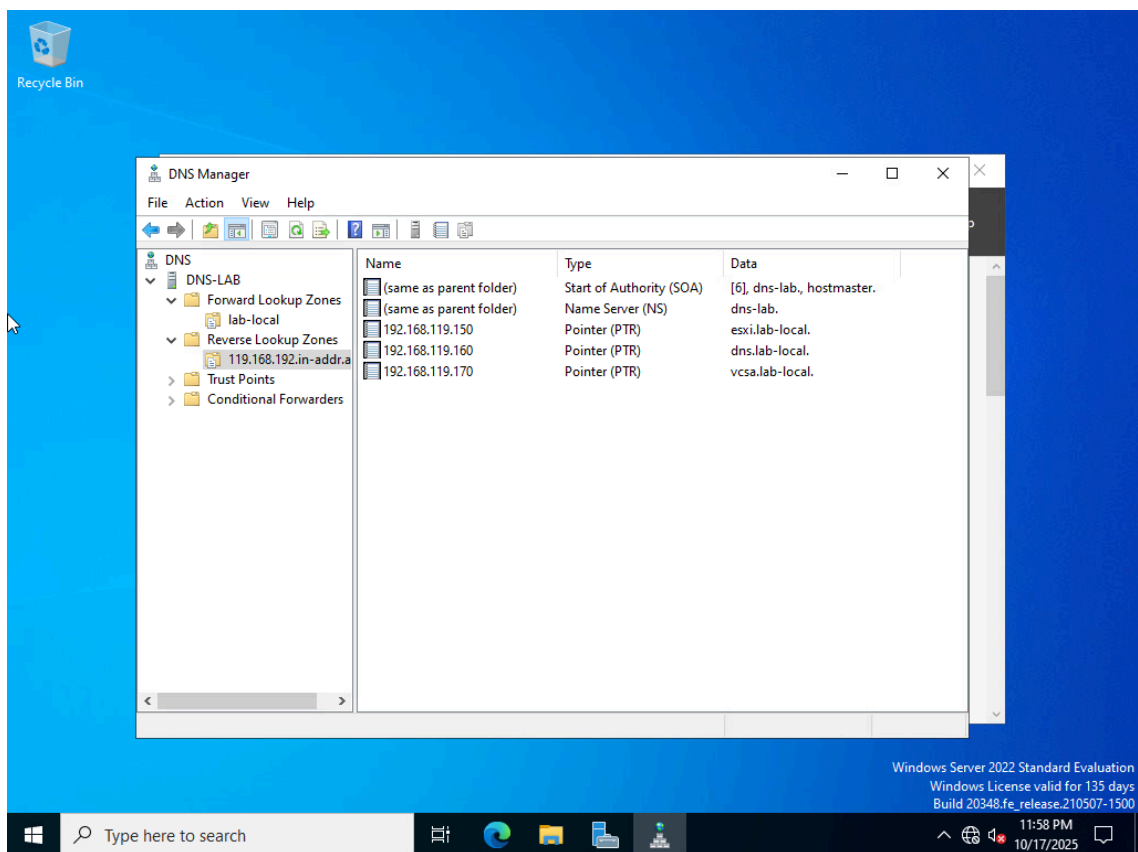
- **Figure 4.3.1 – Server Manager console showing the DNS Server role successfully installed.**



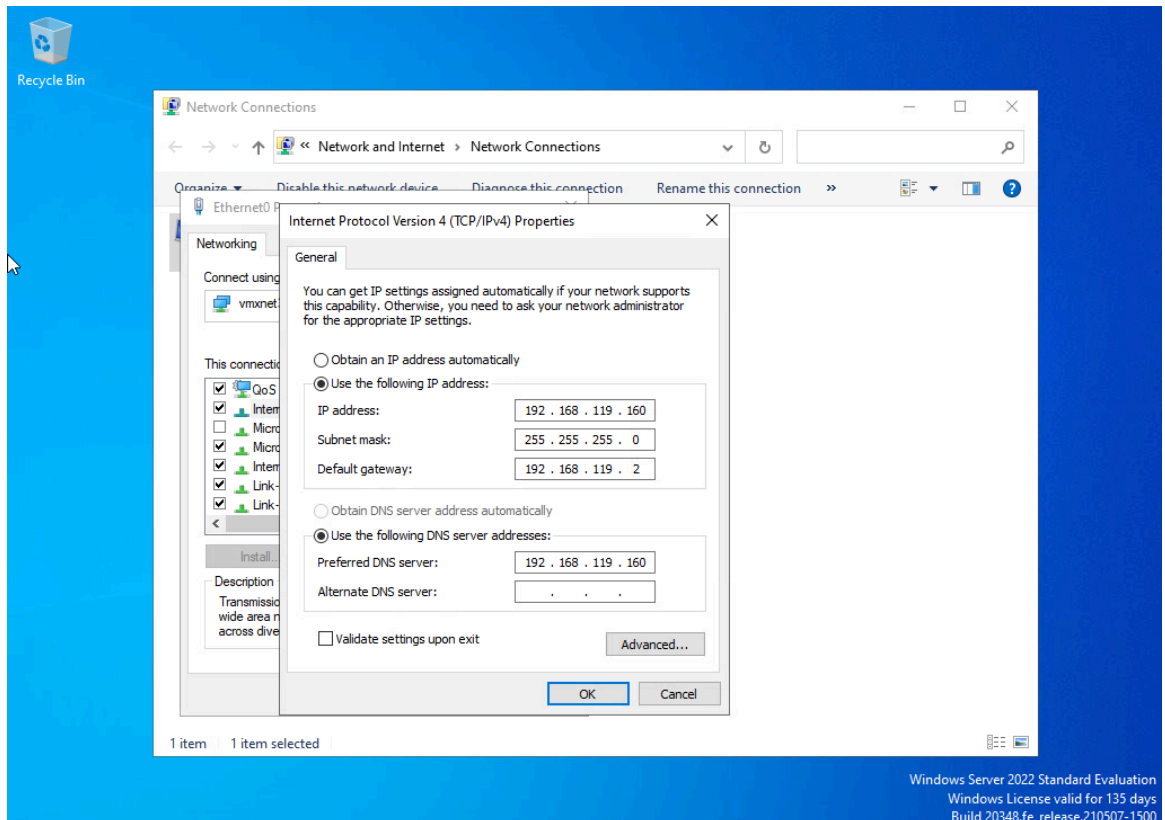
- Figure 4.3.2 – DNS Manager console displaying forward zone **lab.local** and created A records.



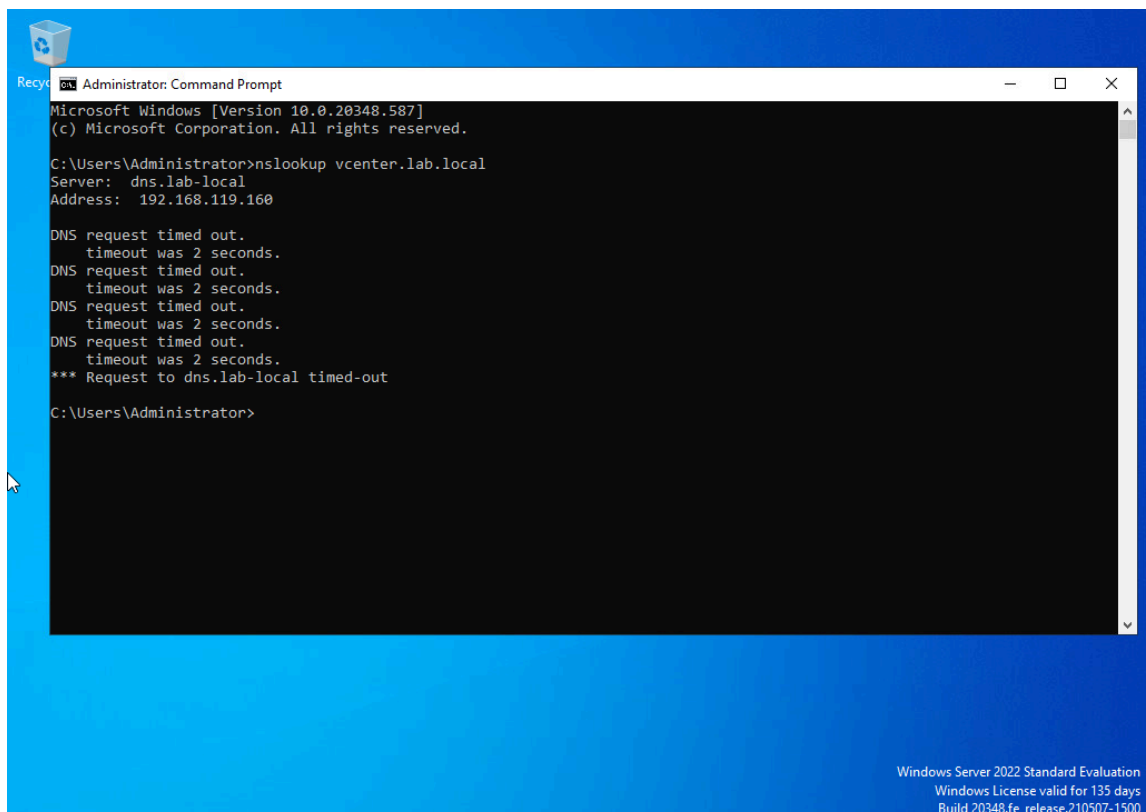
- Figure 4.3.3 – Reverse zone **119.168.192.in-addr.arpa** with associated PTR records.



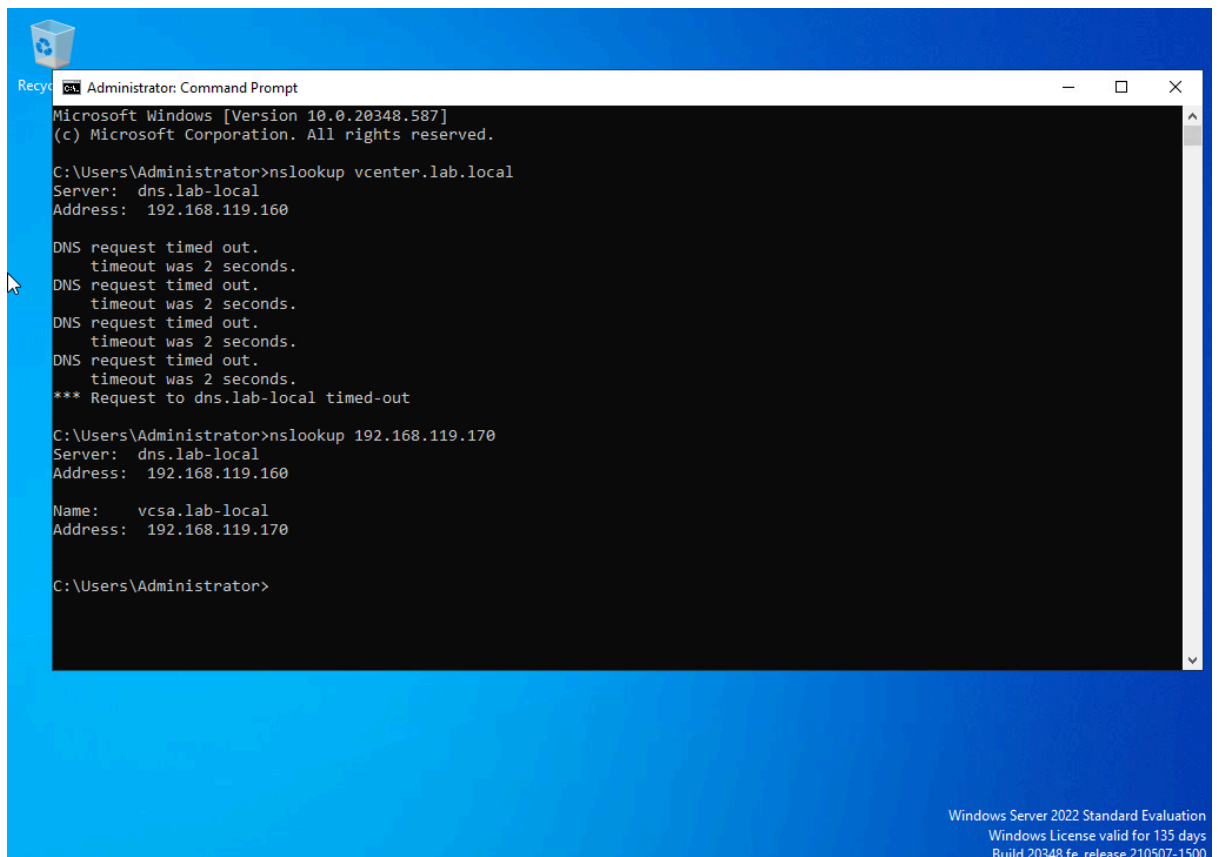
- **Figure 4.3.4 – Network adapter static IP configuration pointing to 192.168.119.160.**



- **Figure 4.3.5 – nslookup vcenter.lab.local test showing correct forward resolution.**



- **Figure 4.3.6 – nslookup 192.168.119.170 test showing reverse resolution.**



```
Administrator: Command Prompt
Microsoft Windows [Version 10.0.20348.587]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Administrator>nslookup vcenter.lab.local
Server: dns.lab-local
Address: 192.168.119.160

DNS request timed out.
    timeout was 2 seconds.
DNS request timed out.
    timeout was 2 seconds.
DNS request timed out.
    timeout was 2 seconds.
DNS request timed out.
    timeout was 2 seconds.
*** Request to dns.lab-local timed-out

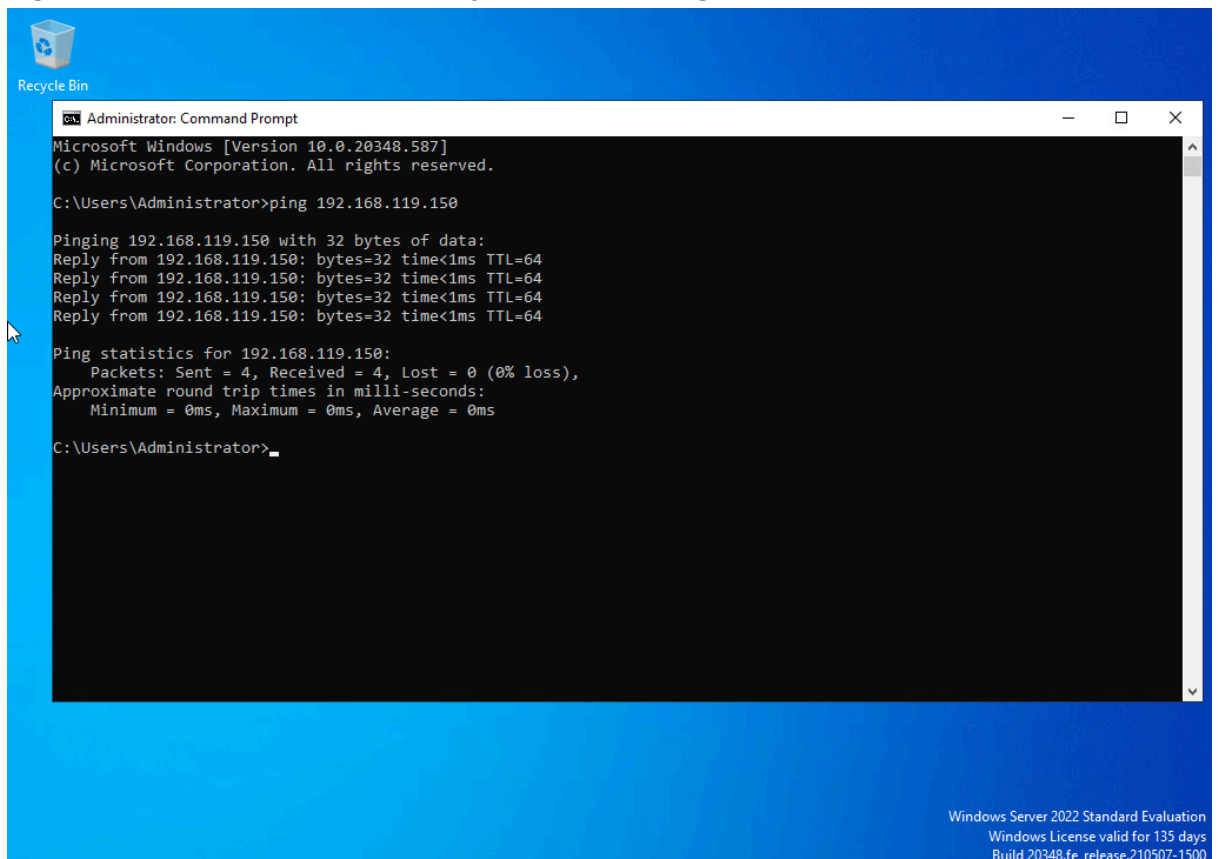
C:\Users\Administrator>nslookup 192.168.119.170
Server: dns.lab-local
Address: 192.168.119.160

Name: vcsa.lab-local
Address: 192.168.119.170

C:\Users\Administrator>
```

Windows Server 2022 Standard Evaluation
Windows License valid for 135 days
Build 20348.fe_release.210507-1500

- **Figure 4.3.7 – General connectivity validation (ping between DNS ↔ ESXi).**



```
Administrator: Command Prompt
Microsoft Windows [Version 10.0.20348.587]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Administrator>ping 192.168.119.150

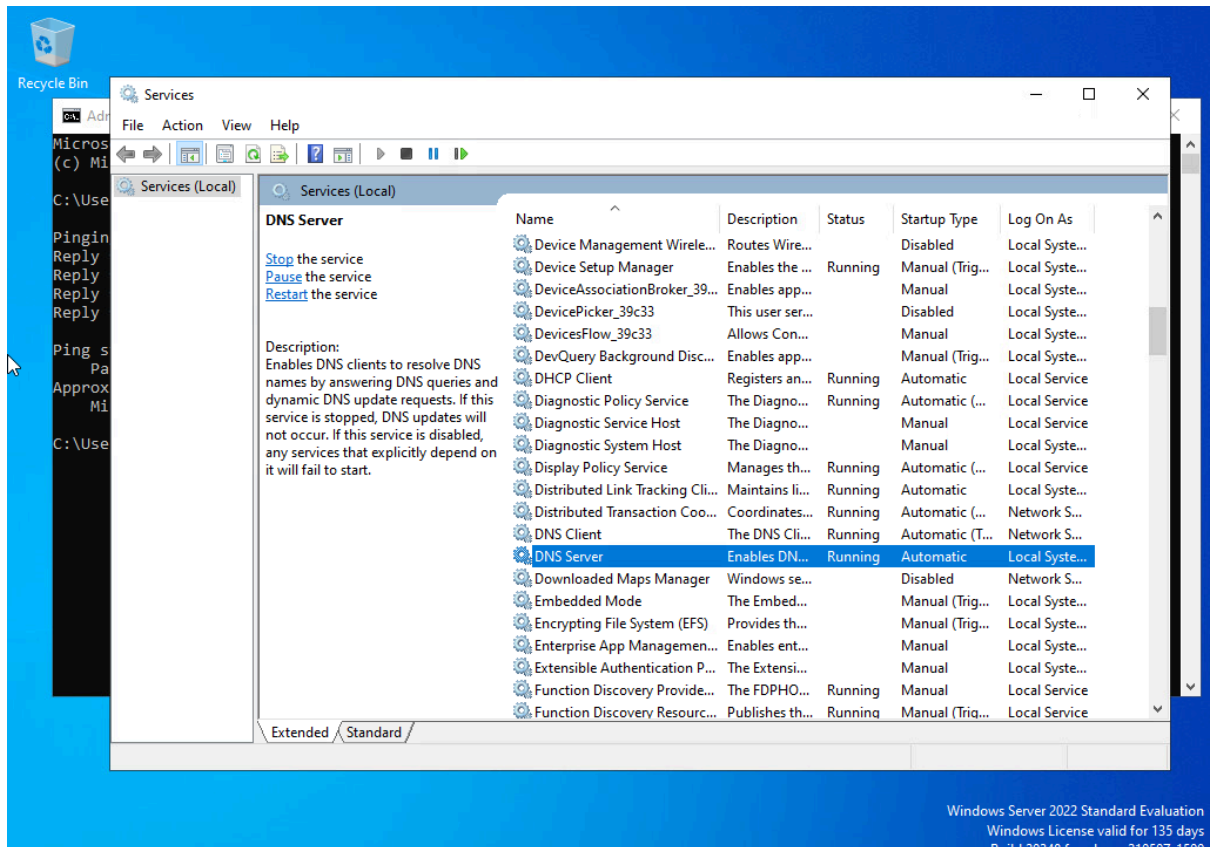
Pinging 192.168.119.150 with 32 bytes of data:
Reply from 192.168.119.150: bytes=32 time<1ms TTL=64
Reply from 192.168.119.150: bytes=32 time<1ms TTL=64
Reply from 192.168.119.150: bytes=32 time<1ms TTL=64
Reply from 192.168.119.150: bytes=32 time<1ms TTL=64

Ping statistics for 192.168.119.150:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

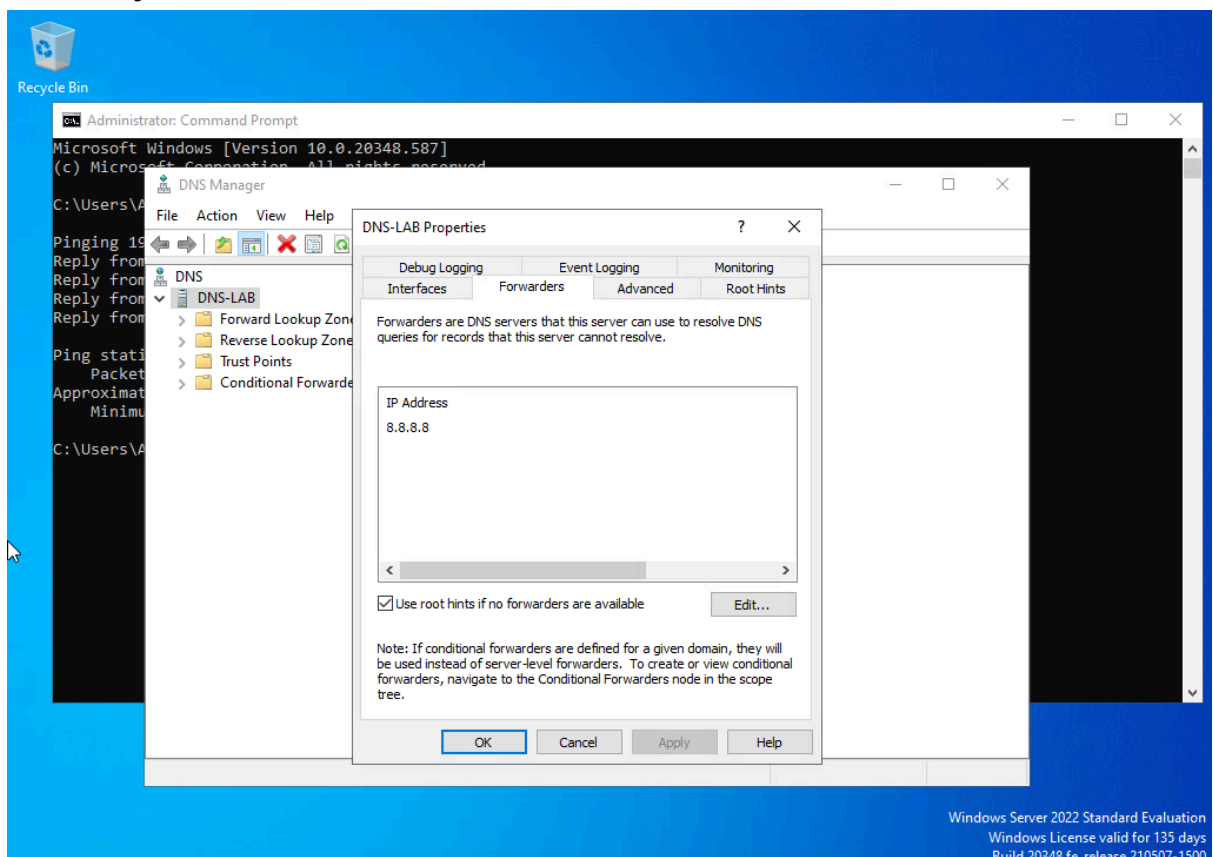
C:\Users\Administrator>
```

Windows Server 2022 Standard Evaluation
Windows License valid for 135 days
Build 20348.fe_release.210507-1500

- Figure 4.3.8 – DNS service running in *Services.msc* (status: Running).



- Figure 4.3.9 – Forwarders configured with 8.8.8.8 (Google Public DNS) as secondary DNS.



4.4 Deployment of the VCSA (vCenter Server Appliance)

4.4.1 Objective

Deploy **vCenter Server Appliance (VCSA) 8.x** on the ESXi host `192.168.119.150`, with static IP `192.168.119.170`, FQDN `vcenter.lab.local`, and internal DNS resolution through the server `192.168.119.160`.

The deployment consisted of two stages:

- **Stage 1:** Appliance installation.
- **Stage 2:** Environment configuration.

Expected services:

- **VAMI:** <https://vcenter.lab.local:5480>
- **vSphere Client:** <https://vcenter.lab.local> or `192.168.119.170`

Active NTP synchronization was configured to prevent time drift between components.

4.4.2 Actions performed

Pre-flight preparation

- Internal DNS operational with A/PTR records:
- `vcenter.lab.local` → `192.168.119.170`
- `esxi.lab.local` → `192.168.119.150`
- `dns.lab.local` → `192.168.119.160`
- Network checks:
- `nslookup vcenter.lab.local 192.168.119.160`
- `nslookup 192.168.119.170 192.168.119.160`
- `ping 192.168.119.170` (no response expected until the VM was created).

- Datastore with more than **600 GB** free (Thin provisioned).
- ESXi host configured with **DNS 192.168.119.160** and **active NTP**.

Attempt 1 — Deployment via UI Installer

- **Source:** Official ISO of VCSA 8.x
([vcsa-ui-installer\win32\installer.exe](#)).
- **Destination:** ESXi host [192.168.119.150](#) (user *root*).
- **Size:** Tiny (2 vCPU / 14 GB RAM).
- **Network:**
 - Port Group: [VMnet8](#) (NAT)
 - IP: [192.168.119.170](#)
 - Netmask: [/24](#)
 - Gateway: [192.168.119.2](#)
 - DNS: [192.168.119.160](#)
 - PNID: [vcenter.lab.local](#)
- **Stage 1:** VM creation and initial boot (progress reached 84–96%).
- **Stage 2:** Assistant start at [:5480](#) — SSO configuration ([vsphere.local](#)), NTP setup, CEIP disabled.

Troubleshooting after freezes during Stage 1

- Reviewed logs on ESXi ([hostd.log](#), [vmkernel.log](#)).
- Verified network and DNS connectivity from ESXi and Windows.
- Checked NTP and resources; repeated tests with 4 vCPU / 16 GB RAM.
- **Result:** Stage 1 completed partially, but **VAMI was not reachable**.

Attempt 2 — Stage 1 completed, Stage 2 stuck (~33%)

- VAMI accessible and wizard launched.
- Long hang on *Starting VMware Security Token Service (STS)*.
- DNS, time, and STS services reviewed from VCSA console.
- Failures correlated with **NTP desynchronization** or **incomplete DNS resolution**.

Attempt 3 — Direct deployment in Workstation (OVF)

- Converted ISO using **OVF Tool** or extracted it via **7-Zip**.
- Edited OVF descriptor (`userConfigurable="true"` when applicable).
- Imported into Workstation with network **VMnet8**, PNID = `vcenter.lab.local`.
- Stage 1 completed successfully; Stage 2 again stuck at STS step, repeating the same DNS/NTP issues.
- Additional tests performed increasing RAM/CPU and validating network.

Cross troubleshooting actions

- **DNS:** Confirmed A + PTR records for *vcenter* and *esxi*; `nslookup` worked correctly from all nodes.
- **NTP:** ESXi synchronized with `es.pool.ntp.org`.
- **Network:** All NICs connected to **VMnet8**; ports **443** and **5480** reachable.
- **Resources:** Minimum for VCSA Tiny (≥ 14 GB RAM / 2 vCPU); tested with **16 GB RAM / 4 vCPU**.
- **Logs reviewed:**
 - Installer client: `%TEMP%\vcsUiInstaller\logs\`
 - ESXi: `/var/log/hostd.log`, `/var/log/vmkernel.log`
 - VCSA: `/var/log/vmware/*`, `vmon-cli`

4.4.3 Issues encountered

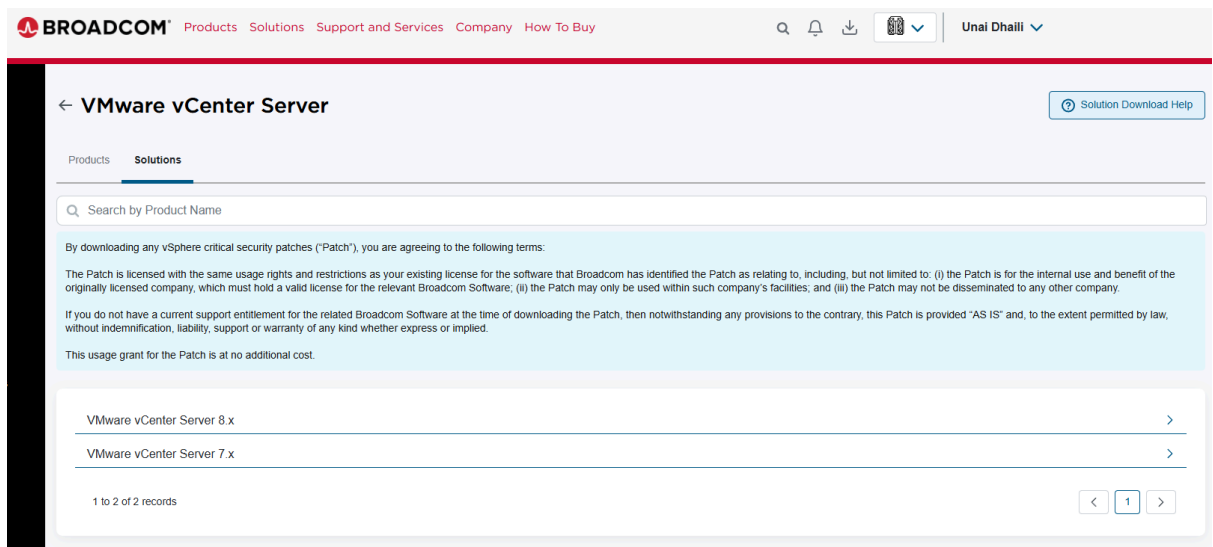
Symptom	Identified cause	Mitigation applied
Stage 1 stuck (84–96%)	Network on wrong segment (not VMnet8), incomplete DNS, or resource exhaustion	Reassign NIC to VMnet8, fix A + PTR records, increase RAM, close host applications
Stage 2 stuck (33% STS/SSO)	Time drift between ESXi and VCSA, missing reverse lookup, inconsistent FQDN	Force NTP on VCSA, validate direct/reverse <code>nslookup</code> , retry Stage 2
OVF error “manifest does not validate”	Incorrect signature or certificate in OVA	Extract with 7-Zip and use OVF Tool with <code>--skipManifestCheck</code>
Physical host freeze (96%)	Lack of RAM/CPU and high disk I/O	Increase VCSA RAM, close heavy processes, ensure datastore free space

4.4.4 Final results

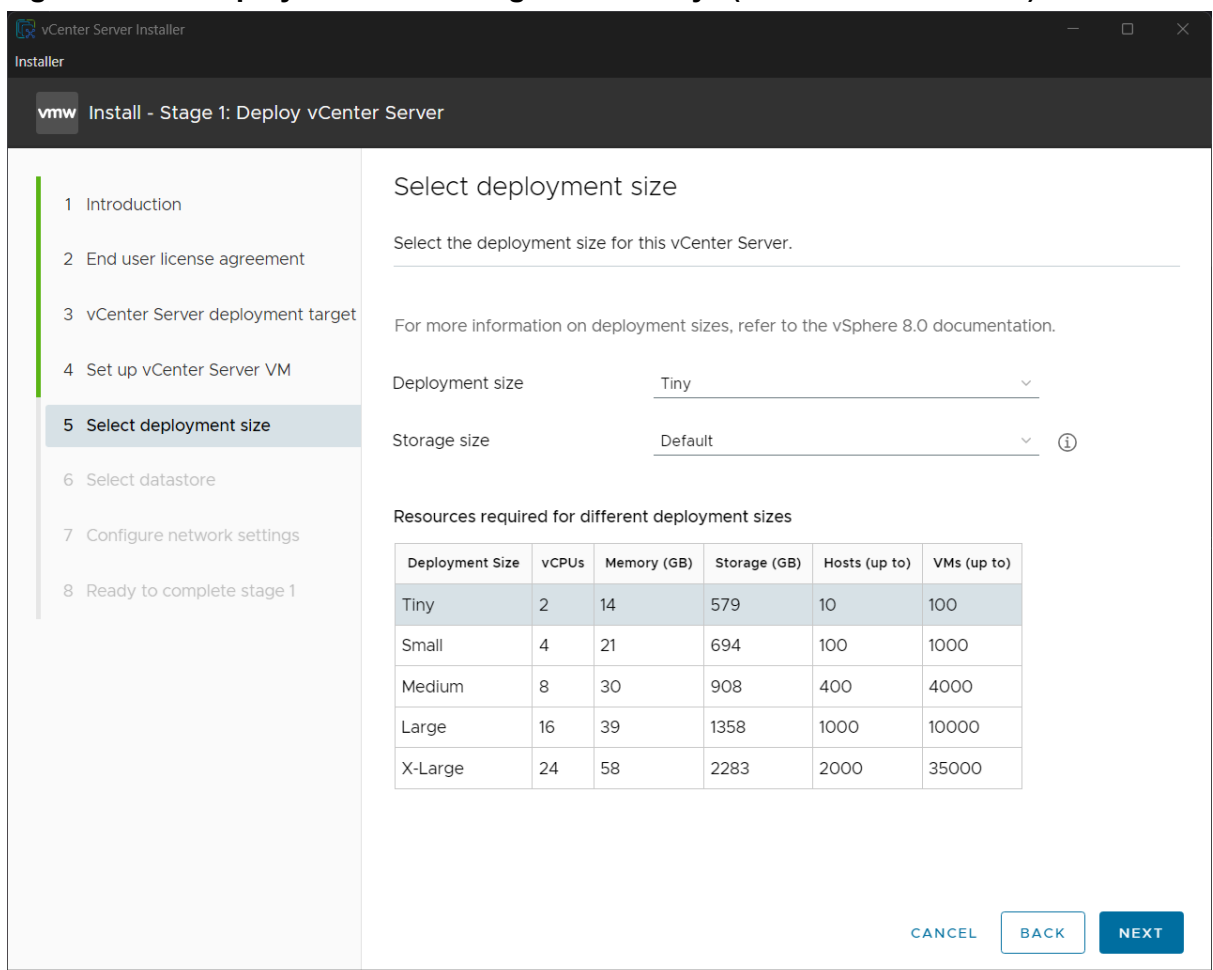
- **VCSA deployment partially successful:** Stage 1 completed several times; Stage 2 started but not finalized due to environment limitations.
- **Stable base infrastructure:** ESXi operational, NTP synchronized, and internal DNS consistent.
- All deployment attempts (UI Installer / OVF Tool) documented along with their troubleshooting.
- **Conclusion:** The failure was due to hardware constraints, not configuration errors.
- Practical experience gained in virtualization, networking, DNS/NTP, and log analysis directly applicable to professional environments.

4.4.5 Evidence

- **Figure 4.4.1 – Official ISO download of VMware vCenter Server Appliance 8.0 from Broadcom portal.**



- **Figure 4.4.2 – Deployment size configuration “Tiny” (2 vCPU / 14 GB RAM).**



- **Figure 4.4.3 – Network configuration: static IP 192.168.119.170, Gateway 192.168.119.2, DNS 192.168.119.160.**

vCenter Server Installer
Installer

vmw Install - Stage 1: Deploy vCenter Server

1 Introduction
2 End user license agreement
3 vCenter Server deployment target
4 Set up vCenter Server VM
5 Select deployment size
6 Select datastore
7 Configure network settings
8 Ready to complete stage 1

vCenter Server deployment target

Specify the vCenter Server deployment target settings. The target is the ESXi host or vCenter Server instance on which the vCenter Server will be deployed.

ESXi host or vCenter Server name	192.168.119.150	①
HTTPS port	443	
User name	root	①
Password		

CANCEL BACK NEXT

- **Figure 4.4.4 – FQDN assignment vcenter.lab.local during setup wizard.**

vCenter Server Installer
Installer

vmw Install - Stage 1: Deploy vCenter Server

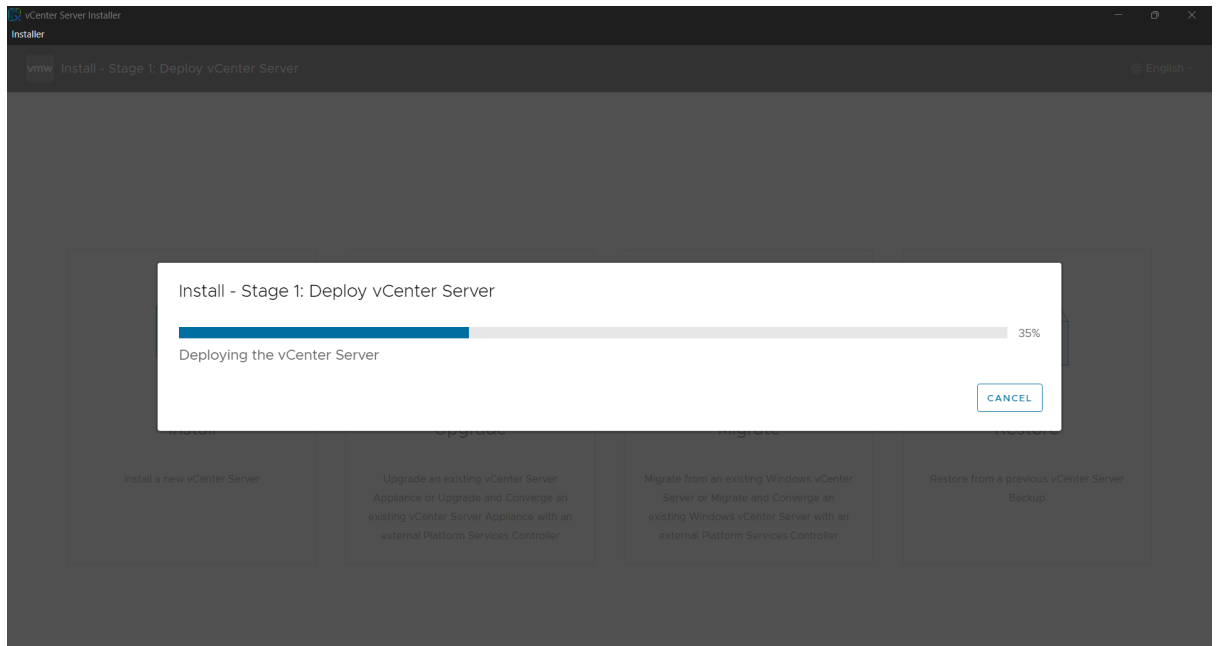
1 Introduction
2 End user license agreement
3 vCenter Server deployment target
4 Set up vCenter Server VM
5 Select deployment size
6 Select datastore
7 Configure network settings
8 Ready to complete stage 1

Configure network settings

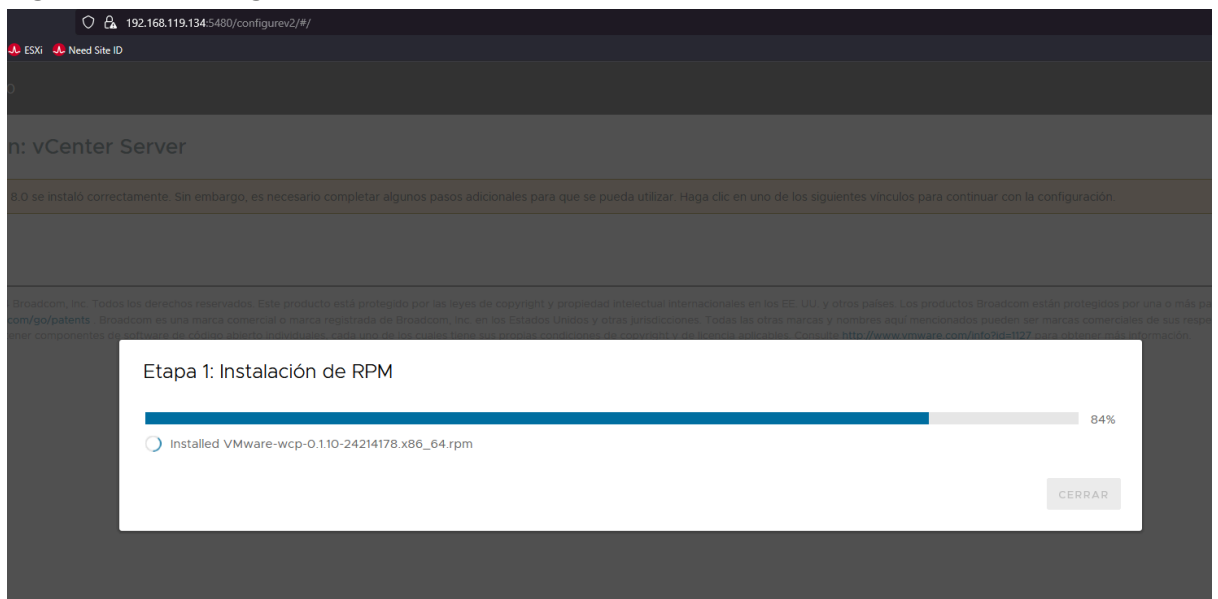
Network	VM Network	①
IP version	IPv4	
IP assignment	static	
FQDN	vcenter.lab.local	①
IP address	192.168.119.170	
Subnet mask or prefix length	255.255.255.0	①
Default gateway	192.168.119.2	
DNS servers	192.168.119.160	
Common Ports		
HTTP	80	
HTTPS	443	

CANCEL BACK NEXT

- **Figure 4.4.5 – Installation process start (Stage 1 – Deploy VCSA).**



- **Figure 4.4.6 – Progress stuck at 84% (initial NIC/network issue).**



- Figure 4.4.7 – Failed ping to **192.168.119.170** during Stage 1 (NIC inactive).

```
C:\Users>ping 192.168.119.170

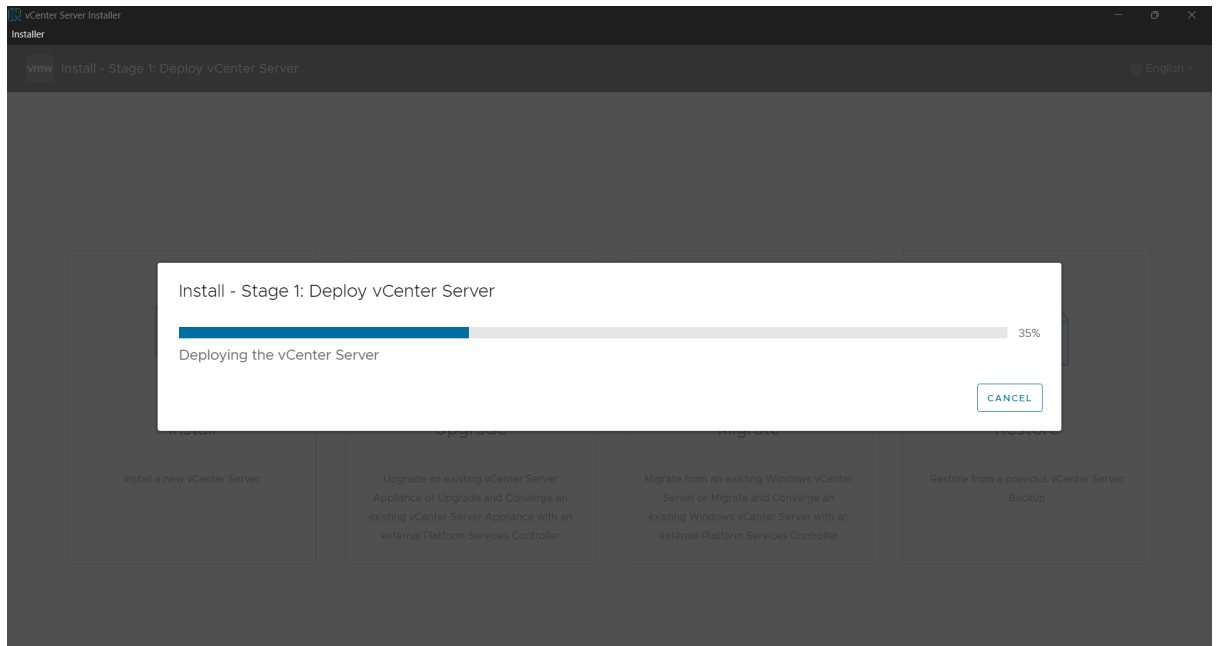
Haciendo ping a 192.168.119.170 con 32 bytes de datos:
Respuesta desde 192.168.119.1: Host de destino inaccesible.
Tiempo de espera agotado para esta solicitud.
Tiempo de espera agotado para esta solicitud.
Tiempo de espera agotado para esta solicitud.

Estadísticas de ping para 192.168.119.170:
    Paquetes: enviados = 4, recibidos = 1, perdidos = 3
              (75% perdidos),

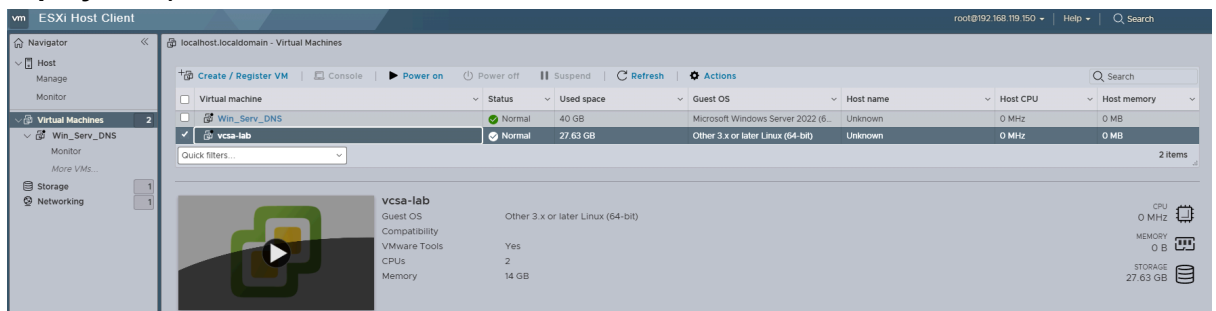
C:\Users>
```

- Figure 4.4.8 – Configuration wizard (Stage 2 – OVF deployment).

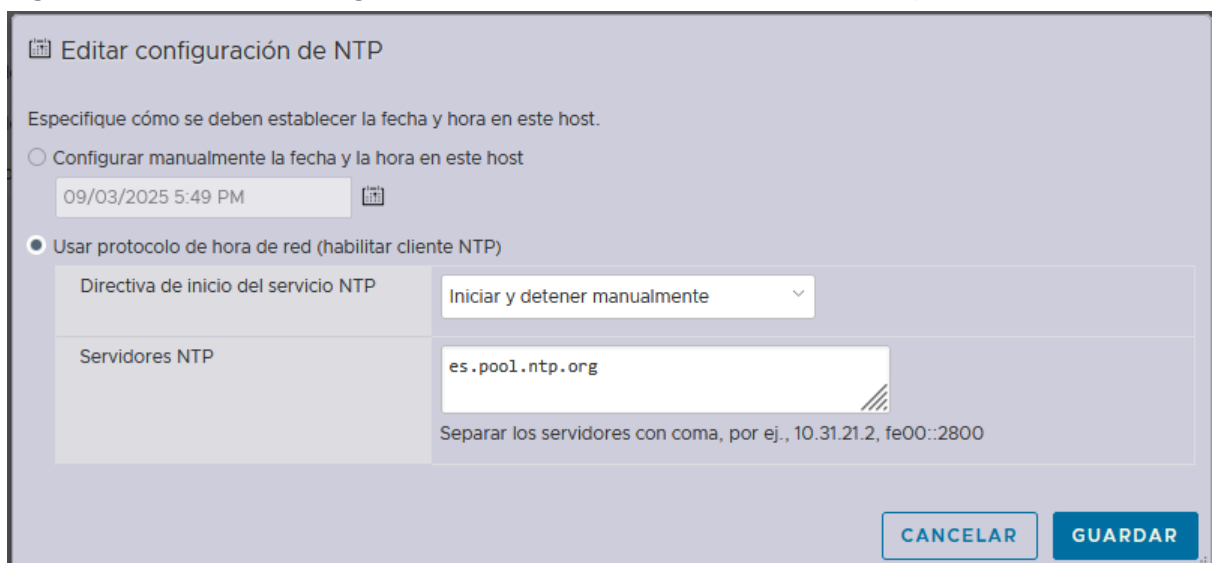
- **Figure 4.4.9 – Hang on Starting VMware Security Token Service (STS) (~35%).**



- **Figure 4.4.10 – VCSA VM created on host but not reachable (incomplete deployment).**



- **Figure 4.4.11 – NTP configuration on ESXi with es.pool.ntp.org.**



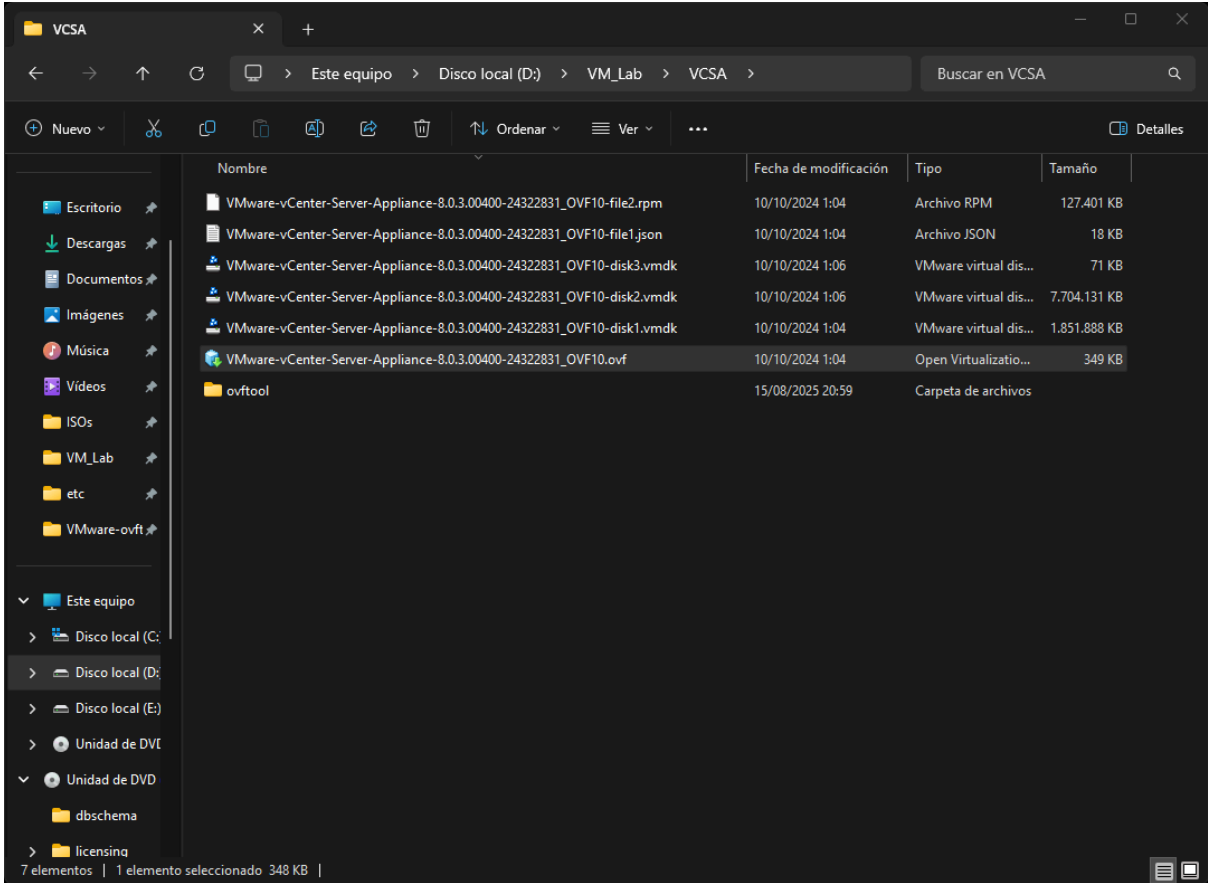
- **Figure 4.4.12 – Manual editing of OVF file in Notepad++** (userConfigurable="true").

```

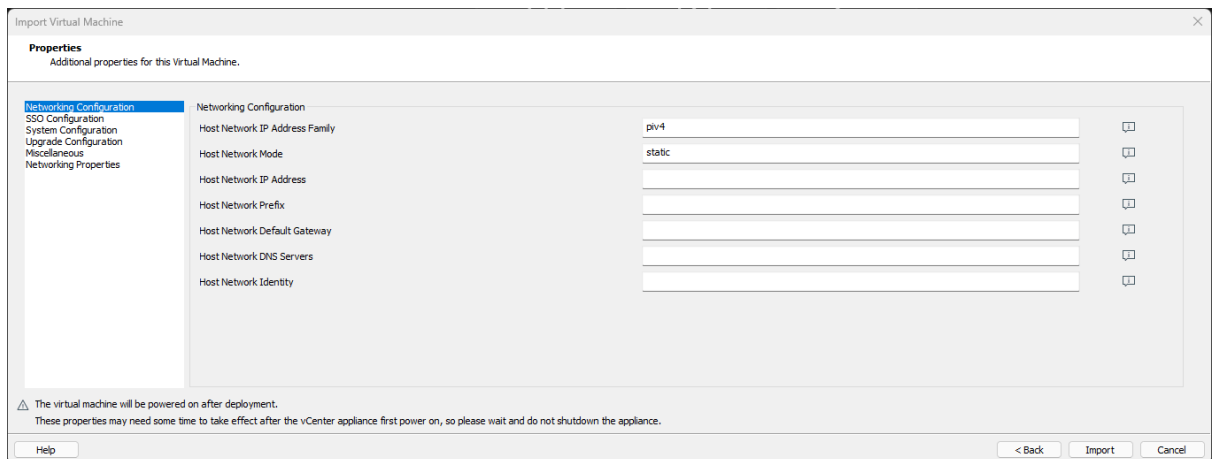
</Property>
<Property ovf:key="guestinfo.cis.upgrade.import.directory" ovf:value="/storage/seat/cis-export-folder">
  <Label>Upgrade Source OS Import Directory</Label>
  <Description>Username for the appliance operating system to upgrade. Usually root. Set only for upgrade.</Description>
</Property>
<Property ovf:key="guestinfo.cis.upgrade.source.guest.password" ovf:password="true" ovf:type="string" ovf:userConfigurable="false" ovf:value="">
  <Label>Upgrade Source OS Password</Label>
  <Description>Password for the appliance operating system to upgrade. Set only for upgrade.</Description>
</Property>
<Property ovf:key="guestinfo.cis.upgrade.source.guestops.host.addr" ovf:type="string" ovf:userConfigurable="false" ovf:value="">
  <Label>Upgrade Management Host Hostname</Label>
  <Description>URL that consists of the IP address or FQDN and https port of the vCenter Server instance on ESXi host that manages the appliance to upgrade. https port is an optional parameter which by default is 443. Example: 10.10.10.10, //10.10.10.10:444, //[2001:db8:a0b:12f0::1]:444. Set only for upgrade.</Description>
</Property>
<Property ovf:key="guestinfo.cis.upgrade.source.guestops.host.user" ovf:type="string" ovf:userConfigurable="false" ovf:value="">
  <Label>Upgrade Management Host Username</Label>
  <Description>Username for the host that manages appliance to upgrade. Can be either vCenter or ESX host. Set only for upgrade.</Description>
</Property>
<Property ovf:key="guestinfo.cis.upgrade.source.guestops.host.password" ovf:password="true" ovf:type="string" ovf:userConfigurable="false" ovf:value="">
  <Label>Upgrade Management Host Password</Label>
  <Description>Password for the host that manages appliance to upgrade. Can be either vCenter or ESX host. Set only for upgrade.</Description>
</Property>
<Property ovf:key="guestinfo.cis.upgrade.source.ssl.thumbprint" ovf:type="string" ovf:userConfigurable="false" ovf:value="">
  <Label>Upgrade Management Host Thumbprint</Label>
  <Description>Thumbprint for the SSL certificate of the host that manages the appliance to upgrade. Set only for upgrade.</Description>
</Property>
<Property ovf:key="guestinfo.cis.upgrade.source.platform" ovf:type="string" ovf:userConfigurable="false" ovf:value="linux">
  <Label>Upgrade Source Platform</Label>
  <Description>Source host platform. Optional. Set only for upgrade.</Description>
</Property>
<Property ovf:key="guestinfo.cis.upgrade.source.export.directory" ovf:type="string" ovf:userConfigurable="false" ovf:value="/var/tmp">
  <Label>Upgrade Source Export Folder</Label>
  <Description>Folder on the source appliance, where to store migrate data. Optional. Set only for upgrade.</Description>
</Property>
<Property ovf:key="guestinfo.cis.upgrade.import.directory" ovf:type="string" ovf:userConfigurable="true" ovf:value="/storage/seat/cis-export-folder">
  <Label>Upgrade Source Import Directory</Label>
  <Description>Folder on the source appliance, where to store migrate data. Optional. Set only for upgrade.</Description>
</Property>

```

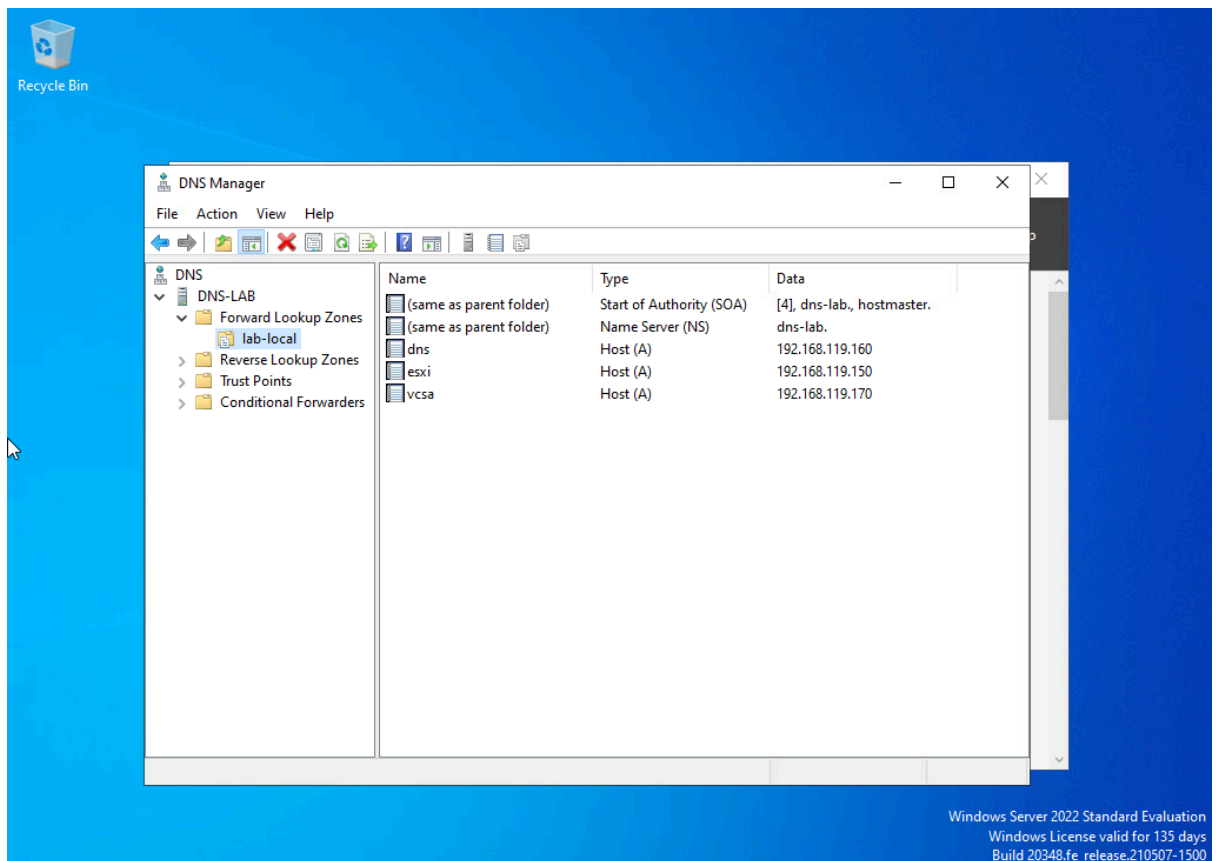
- **Figure 4.4.13 – OVA files extracted: .ovf, .vmdk, and .mf using 7-Zip.**



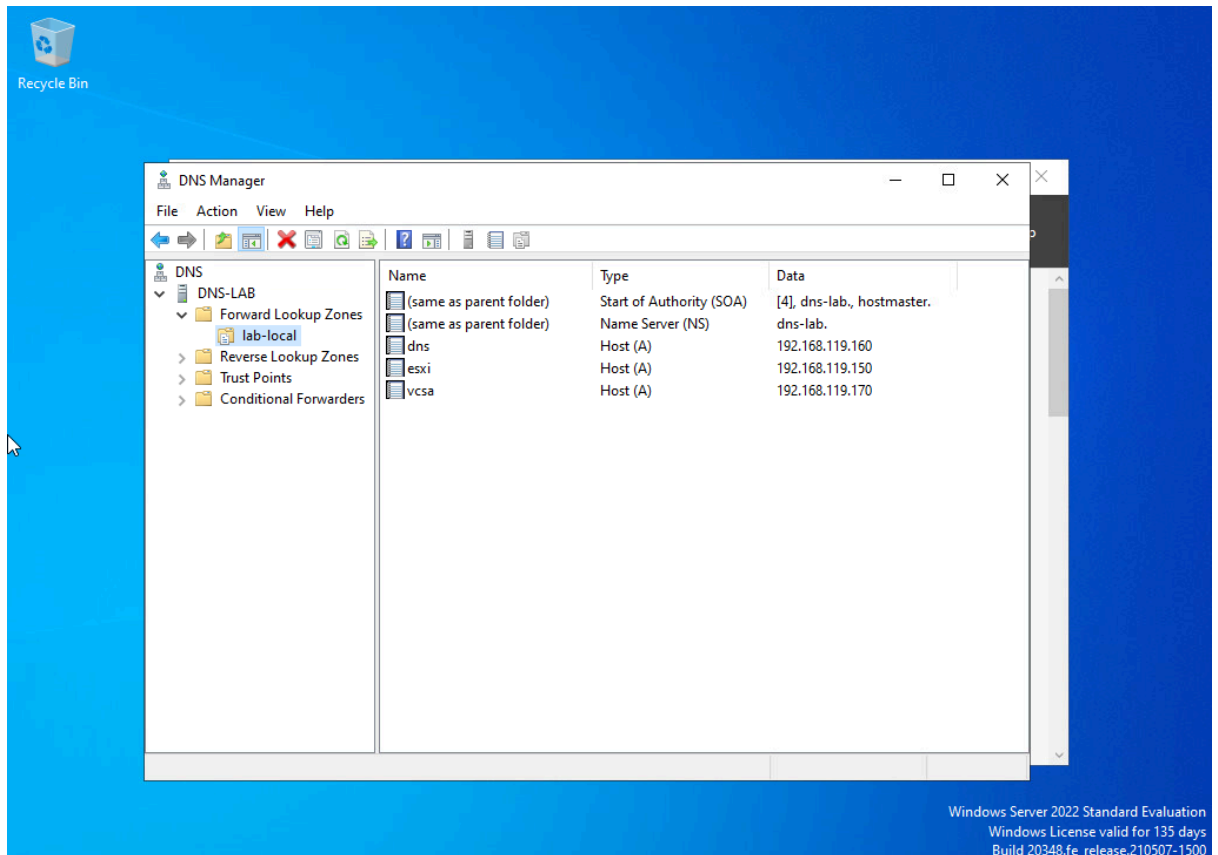
- **Figure 4.4.14 – OVF deployed directly in Workstation (VMnet8, IP 192.168.119.170).**



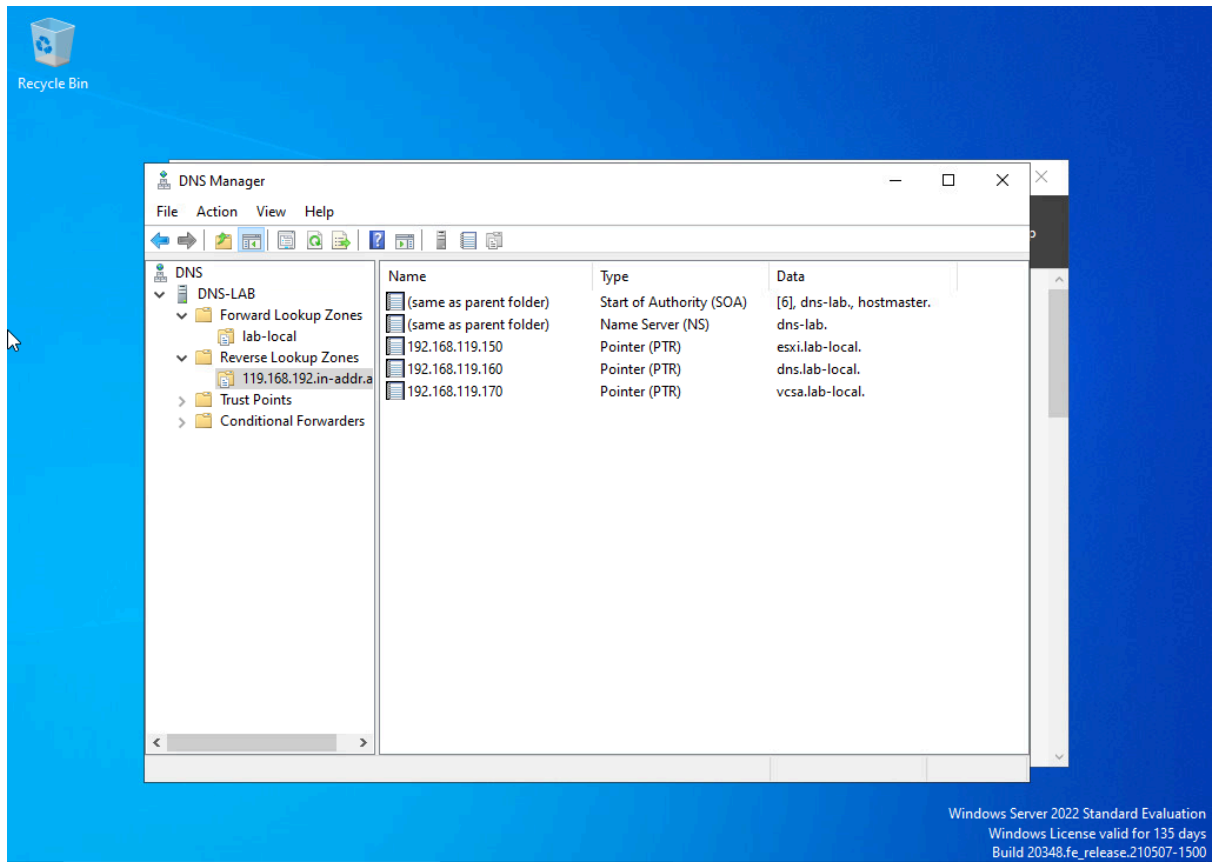
- **Figure 4.4.15 – vmkernel1.log showing ICMP traffic and VCSA network events.**



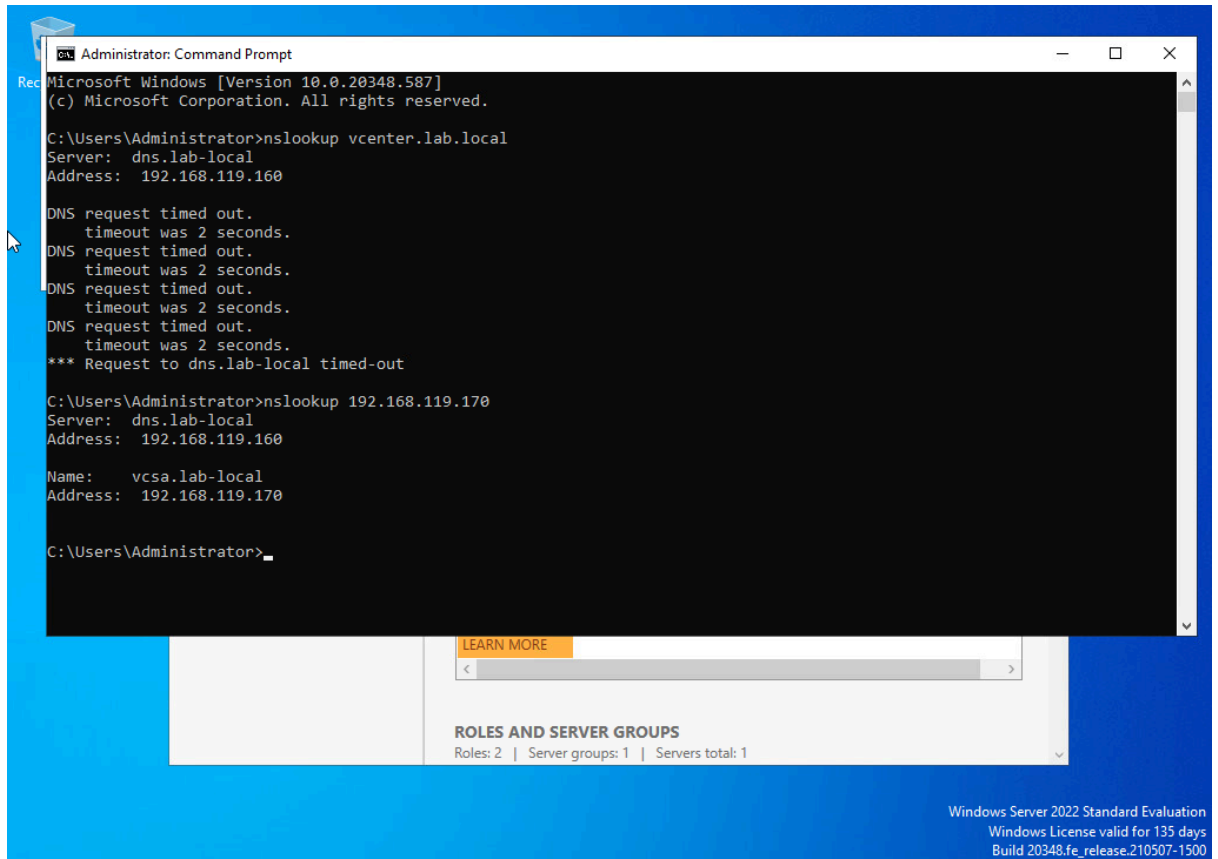
- Figure 4.4.16 – DNS Manager (Windows Server) displaying **lab.local** forward zone.



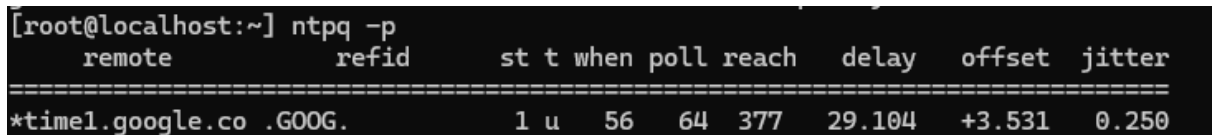
- Figure 4.4.17 – Reverse zone **119.168.192.in-addr.arpa** with correct PTR records.



- **Figure 4.4.18** – **nslookup vcenter.lab.local** and **nslookup 192.168.119.170** from Windows.



- **Figure 4.4.19** – **ntpq -p** output in ESXi showing sync with **time1.google.com**.



5. Conclusions, Lessons Learned, and Potential Improvements

5.1 General Conclusions

The deployment of **vCenter Server Appliance (VCSA)** over a **VMware ESXi 8.0 virtualized environment** represented a high-complexity technical exercise aimed at reproducing a corporate-grade infrastructure within a controlled lab.

Although the full VCSA deployment could not be completed due to environment limitations (resources, compatibility, and stability during Stage 2), the work carried out succeeded in building a solid and fully functional infrastructure around ESXi, NTP, and DNS — significantly deepening the understanding of the internal installation, dependency, and communication processes between vSphere components.

Key achievements:

- Successfully installed and configured an **ESXi 8.0** host, stable and accessible via the web interface.
- Configured the **VMnet8 (NAT)** lab network with consistent static addressing.
- Established **NTP synchronization** with verifiable external servers (time.google.com).
- Implemented an **internal DNS server (Windows Server 2022)** managing both forward and reverse zones ([lab.local](#) and [119.168.192.in-addr.arpa](#)).
- Validated **bidirectional name resolution (A and PTR)** among all components.
- Deployed the VCSA using both the **Graphical Installer (UI Installer)** and **OVF Tool**, analyzing errors and root causes for the Stage 2 failures.
- Thoroughly documented every step, command, log, and test performed.

Even though the complete VCSA deployment was not finalized, the project achieved its main goal: **to reproduce, understand, and document all foundational components of a corporate-grade virtualization environment.**

5.2 Key Lessons Learned

Throughout the project, several essential technical skills were consolidated:

- **vSphere architecture:** deep understanding of the hierarchical relationship between ESXi hosts, vCenter Server, and managed virtual machines, as well as the role of the VCSA as the central management core.
- **DNS and FQDN dependencies:** hands-on validation that forward and reverse name resolution are critical for a successful VCSA deployment and secure ESXi communication.
- **NTP synchronization:** demonstrated that time misalignment between nodes causes service failures in components such as STS or SSO.
- **Network management under NAT (VMnet8):** practical understanding of coherent addressing, gateway configuration, NAT translation, and lab isolation.
- **Advanced troubleshooting:** use of tools such as **OVF Tool**, **journalctl**, **vmon-cli**, **ntpq**, **ping**, and **nslookup** to diagnose complex issues.
- **Technical and mental resilience:** development of systematic thinking and the ability to tackle real-world projects requiring iteration, patience, and continuous improvement.

5.3 Potential Improvements and Future Steps

To complete the project successfully in future iterations, the following adjustments are proposed:

- Increase host resources: at least **4 vCPU** and **16–18 GB RAM** for the Tiny VCSA to ensure Stage 2 stability.
- Check vendor requirements: datastore with **>580 GB free space (Thin provisioned)** and virtual hardware version **≥17**.
- Use a **dedicated DNS** on a bridged network (VMnet8 or VMnet0) to reduce NAT dependency and improve resolution stability.
- Verify **NTP** before deployment, ensuring positive **ntpq -p** output in both ESXi and DNS.
- Evaluate deployment via **CLI Installer (vCSA CLI Installer)** for better control and traceability.
- Test earlier **VCSA versions (7.x)** to rule out compatibility or resource issues.

- Automate lab provisioning using **PowerCLI** or **Bash scripts** to recreate the environment consistently.
- Integrate **Veeam Backup & Replication** once VCSA is fully operational, simulating a complete virtual-machine protection and recovery workflow.

5.4 Final Remarks

The **VMware Virtualization Home Lab** provided a complete technical experience in deploying and managing vSphere environments.

Although the VCSA installation could not be fully completed due to hardware limitations — mainly RAM, stability, and disk I/O — the project achieved a fully operational infrastructure with **ESXi, internal DNS, and active NTP**, leaving a ready-to-deploy foundation for future runs under more capable conditions.

Practically, every step of a real environment was executed: network planning, base service configuration, NTP and DNS validation, and deployment testing via both the graphical installer and OVF Tool.

All errors were identified, documented, and correctly diagnosed, confirming that the failures were **environmental, not conceptual**.

The results demonstrate a complete mastery of the VMware infrastructure workflow, establishing the foundation to successfully replicate the deployment in a professional or shared environment.

Far from being a failure, this lab validates that the **methodology and process were correct and reproducible** — the only limiting factor was the physical hardware.

6. Index of Captures and Final References

6.1 Lab Captures

The captures follow this format:

Figure X.Y.Z – [short descriptive title]

Each figure is placed within its corresponding section, numbered according to the chapter (e.g., **Figure 4.4.15** belongs to section 4.4), and includes a caption explaining the command, result, or interface shown.

6.2 Resources Used

- VMware ESXi 8.0
- VMware Workstation Pro 17
- vCenter Server Appliance 8.0 ISO (Broadcom)
- Windows Server 2022 Evaluation (internal DNS role)
- Windows 11 client as management workstation
- Official VMware 8.0 Documentation — *Installation & Setup Guide*

6.3 Additional Notes

- All tests were performed in an **isolated lab environment**, without access to any corporate network.
- IP addresses, hostnames, and domains ([lab.local](#)) are **simulated for educational purposes**.
- All incidents and results were documented with complete visual evidence.
- The base environment (**ESXi + DNS + NTP**) remained **fully operational**, serving as the foundation for a future successful VCSA deployment.